

Digitalization, Factor-Biased Productivity, and Market Power*

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October 4, 2026

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Abstract

Digitalization may improve labor efficiency, raise productivity across all inputs, or change firms' ability to price above marginal cost. This paper develops a method that sequentially estimates labor-augmenting productivity, Hicks-neutral productivity, and markups. We allow each component to be persistent over time and digitalization to be endogenous. Under perfect competition, we identify the technology parameters using a proxy-variable approach. We then extend the model to firms with market power. This identifies the technology parameters without imposing a particular model of firm competition. We apply the method to Spanish manufacturing firms over 2000–2019 using a multidimensional index of digitalization. Under perfect competition, a 0.10-point increase in the index is associated with 0.37% higher labor-augmenting productivity and 0.62% higher Hicks-neutral productivity in the following period. Under imperfect competition, the corresponding estimates are 0.55% and 0.43%, together with a 0.07% increase in markups. The labor-augmenting response is significantly larger for small firms under both specifications. The markup response is significantly smaller in highly digitalized industries. These results show that digitalization cannot be summarized by a single productivity measure. Separating labor-augmenting productivity, Hicks-neutral productivity, and markups shows whether the estimated relationship operates through production or through the price–marginal-cost wedge.

Keywords: Digitalization, Factor-Biased Productivity, Labor-Augmenting Productivity, Markups, Market Power, Spanish Manufacturing

JEL Classification: D24, L13, L60, O33, J23, C33

*We thank the Department of Economics at Binghamton University for funding the dataset used in this paper. We are also grateful to Shunan Zhao, Ivan Korolev, Chris Hanes, Yingjun Su, and JoonHwan Cho for providing valuable feedback on the earlier version. We especially thank Jordi Jaumandreu for directing us to the dataset and for providing guidance on constructing the price indices from price variations.

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1 Introduction

Digital technologies can improve the efficiency of particular inputs, raise overall productive efficiency, or change firms’ ability to price above marginal cost. These channels have different implications for firms, workers, and consumers. Distinguishing them is therefore essential to understanding the economic consequences of digitalization. Industry surveys indicate that managers increasingly view digital adoption as a means of improving operating efficiency and financial performance (PwC, 2026; Gartner, 2026).

This paper studies how digitalization is associated with the evolution of labor-augmenting productivity, Hicks-neutral productivity, and firm markups. We use panel data on Spanish manufacturing firms from the Encuesta Sobre Estrategias Empresariales (ESEE) covering 2000–2019. We compare a perfect-competition benchmark with a specification that allows firms to have market power. To examine differences across technologies, we construct a 13-variable digitalization index and four component indices: IT technologies, digital human capital, automation, and stakeholder-facing technologies.

Our findings show positive relationships between digitalization and both productivity components, alongside a small markup response. Under imperfect competition, a 0.10-point increase in the digitalization index is associated with approximately 0.55% higher labor-augmenting productivity, 0.43% higher Hicks-neutral productivity, and 0.07% higher markups in the following period, conditional on their previous levels. The overall productivity relationships remain positive and significant under perfect competition. Under both specifications, the labor-augmenting response is significantly larger among small firms than among large firms. Higher adoption levels among large firms therefore need not imply larger incremental labor-efficiency gains.

These results extend the positive digitalization–productivity relationship documented by Gal et al. (2019) by distinguishing labor-augmenting improvements from Hicks-neutral gains. They also qualify the interpretation of the higher markups in digitally intensive industries reported by Calligaris et al. (2018). In our estimates, additional firm digitalization has a significantly smaller markup response in high-digital-intensity industries than in low-digital-intensity industries. The studies examine different objects. Higher industry markup levels need not imply a larger response to further adoption. Our empirical contribution is to separate these responses and show how they vary across firms and technologies.

Identifying these relationships requires separating technology from market power. Much of the digitalization literature measures total factor productivity while treating technological change as Hicks-neutral. This can affect improvements in the efficiency of particular inputs. Omitting factor-biased productivity can also distort output elasticities and production-based markup estimates (Doraszelski and Jaumandreu, 2019; Demirer, 2025). This matters for approaches such as De Loecker and Warzynski (2012), which recover markups using output elasticities and revenue expenditure shares. Incorrectly specified elasticities can lead researchers to attribute technological changes to changes in market power.

Our methodological contribution is a sequential estimator that identifies technology before recovering markup dynamics. We first identify the restricted translog technology parameters from the quantity side. Revenue shares then provide a firm-year markup signal. The state-space step produces the smoothed

productivity and markup summaries reported below, conditional on additional covariance restrictions.

The findings also inform the evaluation of digital adoption policies. The stronger labor-augmenting response among small firms motivates examining support that improves their access to digital technologies and relevant skills. The component estimates identify IT technologies, digital human capital, and automation as relevant areas for such evaluations. Potential benefits include greater labor efficiency and productive capacity. Establishing whether these gains reach workers and consumers requires evidence on wages, employment, prices, and adoption costs alongside productivity and markups. Our estimates provide quantitative benchmarks for these assessments, while causal returns to specific interventions require additional evidence.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and digitalization measures. Section 4 presents the model. Section 5 sets out identification and estimation. Section 6 reports the empirical results, Section 7 examines diagnostics and robustness, and Section 8 discusses policy implications. Section 9 concludes.

2 Related Literature

The paper connects four related literatures: digitalization and productivity, production function estimation, factor-biased technology, and firm-level markups. The main empirical literature on digitalization generally studies total factor productivity, labor productivity, or other aggregate outcomes. A large group of studies reports positive productivity effects of ICT and digital adoption (Nucci et al., 2023; Cardona et al., 2013; Brynjolfsson and Yang, 1996; Brynjolfsson and Hitt, 2000; Baily, 2002; Dedrick et al., 2003; Melville et al., 2004; Stiroh, 2005; Draca et al., 2008; Holt and Jamison, 2009). Other studies find weak or insignificant effects (Acemoglu et al., 2014; DeStefano et al., 2018; Haller and Lyons, 2015; Bertschek et al., 2015; Colombo et al., 2013). For example, Acemoglu et al. (2014) find little evidence that digital intensity raises labor productivity outside the computer-producing sector of U.S. manufacturing.

The empirical methods in this literature also differ. Nucci et al. (2023) use difference-in-differences and propensity-score matching. Gal et al. (2019) use a control-function approach. Anderton et al. (2023) study a panel of EU countries, while Cusolito (2020) use firm-level data from developing countries. These studies are informative about the overall productivity effect of digitalization. They generally do not separate labor-augmenting productivity from Hicks-neutral productivity.

That distinction is central for us. Production-function methods that recover firm-level markups also depend on reliable output elasticities. If technology changes along more than one dimension, an aggregate productivity residual may mix distinct technological channels. The same problem can affect markup measurement because revenue shares reflect both technology and market power. Our framework therefore treats factor-biased productivity and markups jointly rather than interpreting either from a single residual.

This paper also contributes to the literature on production-based markup estimation. A widely used approach is developed by De Loecker and Warzynski (2012), who recover markups from firms' cost-

minimization conditions using production data and at least one freely variable input. Their approach relates the markup to the ratio of an input’s output elasticity to its revenue expenditure share, with an adjustment for unanticipated output shocks, without specifying the form of competition in the product market. The reliability of the resulting markup estimates nevertheless depends on the identification and specification of output elasticities. Doraszelski and Jaumandreu (2019) highlight two difficulties. Unobserved markups can complicate productivity-proxy inversion under IPC, and conventional production-function specifications may fail to reconcile observed input expenditure shares with the implied output elasticities. Similarly, Demirer (2025) shows that neglecting labor-augmenting productivity can distort estimated output elasticities and, consequently, markups. In relation, Raval (2023) finds that markups estimated using materials are negatively correlated with those estimated using labor and shows that non-neutral productivity differences across firms can explain this discrepancy. Flynn et al. (2019) show that production data alone do not identify markups without additional restrictions.

To address these concerns, Doraszelski and Jaumandreu (2019) combine cost-minimization conditions with a restricted translog technology that accommodates both labor-augmenting and Hicks-neutral productivity. They consider a two-stage procedure that uses the lagged inverted production function to control for Hicks-neutral productivity, as well as a simultaneous estimator of technology and markups. Our IPC model separates factor-biased productivity from market power using a distinct identification strategy. We first identify the technology block from the quantity side and then use the revenue-share equation to obtain a firm-year markup signal. Under additional covariance restrictions, Kalman filtering and smoothing produce firm-year predictions of markup and Hicks-neutral productivity.

Other studies address markup identification through restrictions on returns to scale or alternative estimation frameworks. Flynn et al. (2019) show that prior knowledge of returns to scale can resolve the identification problem in production-based markup estimation. When external information on returns to scale is unavailable, they propose imposing constant returns as a practical identifying restriction. Using a different approach, Kumbhakar et al. (2012) develop a stochastic-frontier framework for estimating market power without requiring input-price data. Their framework accommodates estimation both with and without imposing constant returns to scale. Together, these studies highlight the importance of specifying the technological and statistical restrictions that distinguish market power from production characteristics.

Our analysis of digitalization and markups is closely related to Calligaris et al. (2018), who examine firm-level markups across 26 countries over 2001–2014. They find that markups are higher in digitally intensive sectors and that the gap between more and less digitally intensive sectors widened over this period. Their estimates follow the production-based approach of De Loecker and Warzynski (2012), which relies on correctly identified output elasticities. Subsequent research shows that omitting factor-biased productivity can distort these elasticities and the resulting markup estimates (Doraszelski and Jaumandreu, 2019; Demirer, 2025). This markup estimation strategy can be sensitive to misspecification of factor-biased productivity. We address this issue by embedding both labor-augmenting and Hicks-neutral productivity into the production framework. This allows us to jointly examine how firm-level digitalization drives the

evolution of heterogeneous productivity components and firm markups.

Measurement of digitalization creates a second challenge. Many studies use one indicator, such as ICT spending or IT capital, to represent digital transformation (Bresnahan et al., 2002; Brynjolfsson and Hitt, 1996; Bertschek and Kaiser, 2004; Hempell, 2005; Baily and Lawrence, 2001; Stiroh, 1998, 2002; Morrison, 1997; Gordon, 2000; Jorgenson and Stiroh, 1999; Oliner and Sichel, 2000; Jorgenson, 2001). Earlier work notes the limitations of such measures (Black and Lynch, 2001; Greenan and Mairesse, 2000; Loveman, 1994). The ESEE allows a broader approach. Following Añón Higón and Bonvin (2024), we combine 13 variables into an overall index and four component indices. The components capture IT technologies, digital human capital, automation, and interaction with stakeholders.

3 Data

We use the Encuesta Sobre Estrategias Empresariales (ESEE), an annual panel of Spanish manufacturing firms covering 2000–2019. The survey reports output, inputs, digital technology variables, changes in input and output prices, and other firm characteristics. After cleaning and applying the lag requirements, the overall PC estimation sample with the overall index contains 2,213 firms and 17,087 firm-year observations. We retain firms with at least three usable observations in the relevant estimation sample. Sample size varies across specifications because of lag requirements and data availability. We classify firms with fewer than 200 workers as small and firms with at least 200 workers as large. Table 2 reports the corresponding numbers of observations and firms.

The ESEE reports changes in output and materials prices rather than price levels. Following Jau-mandreu and Lin (2018), we construct Paasche price indices for each firm. Separate indices are constructed for output and intermediate inputs. We use these indices to convert nominal production and intermediate input expenditures into real quantities.

Real output is nominal production of goods and services divided by the firm-specific output price index. We construct real capital using the perpetual inventory method with annual depreciation $d_K = 0.15$. With PI_t equal to the investment-price index divided by 100, we have

$$K_{it} = (1 - d_K)K_{i,t-1} + \frac{I_{i,t-1}}{PI_t}.$$

Investment made in period $t - 1$ enters capital used in period t ; we deflate lagged investment by the current-year investment-price index. At the start of each available consecutive-year spell, we initialize capital using positive reported real net capital when available.

Materials are measured by total intermediate consumption. This includes raw materials, energy, and purchased services. We deflate nominal expenditures by the firm-specific intermediate input price index. The resulting measure captures changes in real materials rather than changes in expenditure caused only by prices.

Labor is measured by total hours worked. The hourly wage equals total labor compensation divided

by total hours. This separates the quantity of labor from the price of labor services. We do not use the wage as a measure of labor efficiency. Labor efficiency is recovered from the model under the maintained identification assumptions.

3.1 Digitalization Indices

We construct a multidimensional digitalization index following Añón Higón and Bonvin (2024). The index contains 13 variables grouped into four components. The IT technology component includes investment in computer equipment and the use of external software programming services. The digital human capital component measures external expenditure on employee training in computing and information technologies. The automation component covers the use of industrial robots, computer-aided design, flexible manufacturing systems, local area networks, and CNC machine tools. The stakeholder interaction component includes ownership of an internet domain, a company webpage, online sales to businesses and consumers, and purchases from suppliers through the internet. We include CNC machine tools to reflect the ESEE manufacturing setting.

We construct each digitalization measure according to the type of survey variable. Monetary variables are converted to real values, while categorical variables are converted into indicator variables. The two monetary variables are investment in computer equipment and external expenditure on IT training. The ESEE reports computer equipment investment as a percentage of total capital goods investment. We convert this percentage into an absolute flow by multiplying it by total capital goods investment. We then convert both monetary flows into stocks using the perpetual inventory method with a 15% depreciation rate. Each stock is divided by its industry and year mean. We assign the resulting relative stock to a decile and rescale it as $(d - 1)/9$, where $d \in \{1, \dots, 10\}$. The resulting scores lie between zero and one.

The categorical variables are survey adoption codes rather than ordered measures of intensity. Six variables are observed every four years: robots, CAD, flexible manufacturing systems, LAN, CNC machine tools, and external software programming services. After converting these variables to binary indicators, we carry an observed response forward for at most three years within a firm. This fills the years between survey waves. It does not replace an explicit missing response or nonresponse code. The five stakeholder variables are observed annually and remain binary after recoding.

The overall index is the unweighted sum of the 13 transformed variables. At the final aggregation stage, we set a missing transformed component to zero. We then normalize the raw sum to a scale from zero to one using its minimum and maximum values. A higher value indicates a broader and more intensive digital presence. We construct the four component indices in the same way.

Finally, we classify firms as either highly or low digitalized based on the taxonomy proposed by Calvino et al. (2018).¹ We further stratify our sample by size, defining small firms as those with fewer than 200 employees, and large firms as those with 200 employees or more. We conduct our estimations on these

¹Following Calvino et al. (2018), low digitalized firms are those operating in the metals and metal products, non-metallic minerals, chemical products, food, beverages and tobacco, and textile, leather, and footwear industries. All remaining firms are classified as highly digitalized.

distinct sub-samples as well as on the pooled full sample.

3.2 Summary Statistics

Table 1 presents summary statistics for output, inputs, and prices across 17,087 firm-year observations. Mean real gross output is 51.3 million euros, compared with a median of 10.1 million euros and a standard deviation of 102.9 million euros. Real capital and materials average 12.4 million and 36.2 million euros, respectively, with corresponding medians of 2.0 million and 6.5 million euros. Annual labor input averages approximately 314 thousand hours per firm, with a median of 101 thousand hours and a standard deviation of 483.5 thousand hours. Across these variables, means substantially exceed medians, and standard deviations exceed means, indicating considerable heterogeneity in production scale and suggesting right-skewed distributions.

Price indices show less relative dispersion than output and input quantities. The output price index averages 1.008, with a standard deviation of 0.094, whereas the materials price index averages 1.038, with a standard deviation of 0.153. Nominal hourly wages average 19.29 euros, with a median of 18.35 euros; the middle half of observations lies between 13.68 and 23.72 euros per hour. The capital investment-price index averages 0.831, with a standard deviation of 0.089. These statistics describe variation across the pooled firm-year sample and therefore reflect both differences across firms and changes over time.

Table 1: Summary statistics for output, inputs, and prices

Variable	Observations	Mean	SD	P25	Median	P75
Real gross output (million)	17,087	51.326	102.912	2.396	10.083	47.380
Real capital (million)	17,087	12.412	25.570	0.433	2.042	11.004
Labor (thousand hours)	17,087	313.972	483.489	41.000	101.000	378.833
Real materials (million)	17,087	36.206	74.528	1.400	6.495	32.338
Output price index	17,087	1.008	0.094	0.953	0.996	1.051
Materials price index	17,087	1.038	0.153	0.948	1.014	1.112
Nominal hourly wage (hour)	17,087	19.287	7.458	13.678	18.347	23.724
Capital investment-price index	17,087	0.831	0.089	0.749	0.862	0.912

Notes: Statistics use the overall PC estimation sample. Real output and materials are nominal values divided by their firm-specific price indices. The hourly wage is total personnel cost divided by total hours. The final row reports the investment-price index for the same observations. Monetary values are displayed in the units indicated in each row.

Table 2 reports summary statistics for the digitalization measures together with sample coverage. The overall estimation with the overall index uses 17,087 firm-year observations from 2,213 firms. The estimation with the four component indices uses 15,874 observations from 2,140 firms. The table reports the corresponding counts for each group. Differences across specifications reflect sample cleaning, data availability, and lag requirements.

Small firms employ fewer than 200 workers, while large firms employ at least 200. The low digital industry group includes metal and metal products, nonmetallic minerals, chemical products, food, drink and tobacco, and textiles, leather and shoes. The high digital group includes agricultural and industrial machinery, electrical goods, transport equipment, timber and furniture, and paper and printing. We ex-

clude other manufacturing from these industry comparisons. These groups classify industries rather than individual firms based on their own digitalization scores.

The mean overall digitalization index is 0.356 in the overall sample. The mean is 0.291 for small firms and 0.526 for large firms. Large firms also have higher means for all four component indices. The largest difference is in digital human capital, with means of 0.151 for small firms and 0.611 for large firms. Differences across industry groups are smaller. The mean overall index is 0.334 in low-digital industries and 0.394 in high-digital industries. These comparisons are descriptive and are based on the relevant estimation samples.

The component scores combine binary adoption measures with transformed expenditure stocks. A zero score for digital human capital does not necessarily imply that a firm provides no training. It may reflect the lowest relative stock category or the treatment of a missing component. The scores therefore measure relative digitalization rather than physical technology stocks or adoption probabilities.

Table 2: Digitalization summary statistics and estimation-sample coverage

Measure	Observations	Firms	Mean	SD	P25	Median	P75
<i>Overall</i>							
Overall index	17,087	2,213	0.356	0.208	0.187	0.346	0.508
IT technologies	15,874	2,140	0.431	0.325	0.143	0.375	0.714
Digital human capital	15,874	2,140	0.276	0.369	0.000	0.000	0.500
Automation	15,874	2,140	0.388	0.324	0.200	0.400	0.600
Stakeholder-facing	15,874	2,140	0.328	0.232	0.200	0.400	0.400
<i>Small firms</i>							
Overall index	12,246	1,689	0.291	0.181	0.154	0.275	0.421
IT technologies	11,389	1,625	0.334	0.284	0.100	0.250	0.556
Digital human capital	11,389	1,625	0.151	0.273	0.000	0.000	0.250
Automation	11,389	1,625	0.314	0.293	0.000	0.200	0.600
Stakeholder-facing	11,389	1,625	0.297	0.224	0.200	0.200	0.400
<i>Large firms</i>							
Overall index	4,343	546	0.526	0.183	0.385	0.529	0.663
IT technologies	3,988	525	0.691	0.282	0.444	0.786	0.944
Digital human capital	3,988	525	0.611	0.385	0.286	0.714	1.000
Automation	3,988	525	0.580	0.323	0.400	0.600	0.800
Stakeholder-facing	3,988	525	0.414	0.247	0.200	0.400	0.600
<i>Low-digital industries</i>							
Overall index	10,610	1,374	0.334	0.204	0.171	0.321	0.485
IT technologies	9,871	1,328	0.425	0.325	0.143	0.375	0.714
Digital human capital	9,871	1,328	0.270	0.368	0.000	0.000	0.500
Automation	9,871	1,328	0.348	0.315	0.000	0.200	0.600
Stakeholder-facing	9,871	1,328	0.315	0.229	0.200	0.400	0.400
<i>High-digital industries</i>							
Overall index	5,883	789	0.394	0.208	0.240	0.387	0.538
IT technologies	5,459	762	0.442	0.325	0.167	0.389	0.750
Digital human capital	5,459	762	0.288	0.367	0.000	0.000	0.600
Automation	5,459	762	0.459	0.327	0.200	0.400	0.800
Stakeholder-facing	5,459	762	0.351	0.236	0.200	0.400	0.400

Notes: Statistics use the PC estimation samples. Digital indices are winsorized within each sample and year at the 2.5th and 97.5th percentiles. Nominal expenditures, price indices, and real quantities follow the definitions in Section 3. Firms may appear in more than one subgroup over time.

4 Model

This section first specifies a production technology with labor-augmenting and Hicks-neutral productivity. We then present the PC benchmark followed by its extension to IPC. Firms are price takers in the labor and materials markets in both cases. Only the output-market assumption changes. The production technology is shared across the two cases. The IPC model additionally requires the dynamic moment and covariance restrictions stated below.

4.1 Technology

Firm i produces gross output at time t using predetermined capital K_{it} , freely variable labor L_{it} , and freely variable materials M_{it} . The production technology is given by

$$Y_{it} = Y_{it}^* e^{\eta_{it}} = F(K_{it}, e^{\omega_{L,it}} L_{it}, M_{it}) e^{\omega_{N,it} + \eta_{it}}, \quad (1)$$

where η_{it} is a productivity shock unobserved by all agents, $\omega_{L,it}$ is labor-augmenting productivity, and $\omega_{N,it}$ is Hicks-neutral productivity. Both $\omega_{N,it}$ and $\omega_{L,it}$ are known to the firm when inputs are chosen, whereas η_{it} is realized after inputs are chosen. We assume $\mathbb{E}[\eta_{it} \mid \mathcal{I}_{it}] = 0$ and define

$$\theta \equiv \mathbb{E}[e^{\eta_{it}} \mid \mathcal{I}_{it}], \quad (2)$$

which is assumed to be constant across firms and over time.

We use the lower-case letters as natural logarithms of their upper-case counterparts, and write $l_{it}^* \equiv l_{it} + \omega_{L,it}$ for effective labor in logs. The firm's information set $\mathcal{I}_{it}$ contains K_{it} , $\omega_{L,it}$, $\omega_{N,it}$, all input prices, and the entire history of these variables.

We assume $\ln F(\cdot)$ to be translog:

$$\begin{aligned} \ln F(K_{it}, L_{it}^*, M_{it}) = & \alpha_0 + \beta_K k_{it} + \beta_L l_{it}^* + \beta_M m_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \frac{1}{2} \beta_{LL} (l_{it}^*)^2 + \frac{1}{2} \beta_{MM} m_{it}^2 \\ & + \beta_{KL} k_{it} l_{it}^* + \beta_{KM} k_{it} m_{it} + \beta_{ML} m_{it} l_{it}^*. \end{aligned} \quad (3)$$

Following Doraszelski and Jaumandreu (2019) and Malikov et al. (2024) (MZZ), we impose separability of capital from the two variable inputs, which requires $\beta_{KL} = \beta_{KM} = 0$. We further impose homogeneity of degree ν in effective labor and materials, which requires $\beta_{LL} + \beta_{ML} = 0$ and $\beta_{MM} + \beta_{ML} = 0$. Denoting $\delta \equiv \beta_{ML}$, these restrictions imply $\beta_{LL} = \beta_{MM} = -\delta$, and we define the degree of homogeneity as

$$\nu \equiv \beta_L + \beta_M. \quad (4)$$

Under these restrictions, the technology can be written as

$$y_{it}^* = \alpha_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \beta_M m_{it} + \beta_L (l_{it} + \omega_{L,it}) - \frac{1}{2} \delta (m_{it} - l_{it} - \omega_{L,it})^2 + \omega_{N,it}. \quad (5)$$

4.2 Output elasticities

Differentiating (5) at fixed $\omega_{L,it}$ gives the observation-specific output elasticities

$$\varepsilon_{it}^L = \beta_L + \delta(m_{it} - l_{it} - \omega_{L,it}), \quad (6)$$

$$\varepsilon_{it}^M = \beta_M - \delta(m_{it} - l_{it} - \omega_{L,it}), \quad (7)$$

$$\varepsilon_{it}^K = \beta_K + \beta_{KK}k_{it}. \quad (8)$$

Homogeneity implies that the elasticity with respect to a proportional change in labor and materials is a constant,

$$\varepsilon_{it}^L + \varepsilon_{it}^M = \nu. \quad (9)$$

4.3 PC model

We use the PC model in which the firm takes the output price P_{it}^Y as exogenously given as the benchmark. Our PC estimation follows the approach of Malikov et al. (2024) (MZZ). Since η affects realized output, the first-order conditions (FOCs) of expected profit maximization over L_{it} and M_{it} imply

$$P_{it}^Y \theta \varepsilon_{it}^L \frac{e^{y_{it}^*}}{L_{it}} = P_{it}^L, \quad (10)$$

$$P_{it}^Y \theta \varepsilon_{it}^M \frac{e^{y_{it}^*}}{M_{it}} = P_{it}^M. \quad (11)$$

We now define the revenue shares and their sum,

$$s_{it}^L = \frac{P_{it}^L L_{it}}{P_{it}^Y Y_{it}}, \quad s_{it}^M = \frac{P_{it}^M M_{it}}{P_{it}^Y Y_{it}}, \quad R_{it} = s_{it}^L + s_{it}^M. \quad (12)$$

Dividing (10) and (11) by revenue $P_{it}^Y Y_{it}$, adding, and using (9) gives the share equation

$$\ln R_{it} = \ln(\theta\nu) - \eta_{it}. \quad (13)$$

Taking the ratio of (10) and (11) gives

$$\frac{P_{it}^L L_{it}}{P_{it}^M M_{it}} = \frac{\varepsilon_{it}^L}{\varepsilon_{it}^M} = \frac{\beta_L + \delta(m_{it} - l_{it} - \omega_{L,it})}{\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})}. \quad (14)$$

The output price, Hicks-neutral productivity, θ , and η_{it} all cancel.

Solving (14) gives

$$\delta(m_{it} - l_{it} - \omega_{L,it}) = \beta_M SC_{it}^L - \beta_L SC_{it}^M = \nu SC_{it}^L - \beta_L, \quad (15)$$

and hence the labor-augmenting-productivity term can be expressed as

$$\omega_{L,it} = m_{it} - l_{it} + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L. \quad (16)$$

where labor share of total variable cost is

$$SC_{it}^L = \frac{P_{it}^L L_{it}}{P_{it}^L L_{it} + P_{it}^M M_{it}}, \Rightarrow SC_{it}^M = 1 - SC_{it}^L. \quad (17)$$

Substituting (15) into (6)–(7) yields the identities

$$\varepsilon_{it}^L = \nu SC_{it}^L, \quad \varepsilon_{it}^M = \nu SC_{it}^M. \quad (18)$$

Equations (16) and (18) are implications of the restricted translog with two variable inputs. They hold in both market environments studied below, because a markup common to the two conditions cancels from the ratio. This common inversion is the point at which our extension connects to the competitive benchmark.

Substituting (15) into (5) and collecting terms concentrates effective labor out of the production function:

$$y_{it}^* = \kappa_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \nu m_{it} - \frac{\nu^2}{2\delta} (SC_{it}^L)^2 + \omega_{N,it}, \quad \kappa_0 = \alpha_0 + \frac{\beta_L^2}{2\delta}. \quad (19)$$

Equation (19) expresses output in terms of ν , δ , and the composite intercept κ_0 .

To estimate the competitive benchmark model outlined above, we adopt the following three-step procedure.²

4.3.1 Step 1: Estimation of θ , ν and prediction of η_{it}

Let N_{obs} denote the number of firm-years of observations. Because $\mathbb{E}[\eta_{it}] = 0$, the sample analogues of (13) are

$$\hat{\eta}_{it} = \overline{\ln R} - \ln R_{it}, \quad (20)$$

$$\hat{\theta} = \frac{1}{N_{\text{obs}}} \sum_{it} \exp(\hat{\eta}_{it}), \quad (21)$$

$$\hat{\nu} = \frac{\exp(\overline{\ln R})}{\hat{\theta}}. \quad (22)$$

The first step therefore identifies $\nu = \beta_L + \beta_M$ and θ and gives a prediction of η_{it} .

²For details on the estimation of the restricted translog model with factor-biased productivity, see Malikov et al. (2024).

4.3.2 Step 2: Prediction of labor-augmenting productivity

Let $\mathbf{D}_{it}$ collect the firm's digitalization measures. We assume that labor-augmenting productivity follows the following law of motion

$$\omega_{L,it} = \rho_L \omega_{L,i,t-1} + \gamma'_L \mathbf{D}_{i,t-1} + \xi_{L,it}, \quad \mathbb{E}[\xi_{L,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (23)$$

The equation contains no intercept. This is standard in the IO productivity literature (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015; Zhao et al., 2024)

Step 2 estimates the law of motion in (23). Since $\omega_{L,it}$ is unobserved, we replace it using the labor productivity proxy in (16). Substituting the proxy at t and $t - 1$ gives

$$\begin{aligned} \xi_{L,it} = & (m_{it} - l_{it}) + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L \\ & - \rho_L \left[(m_{i,t-1} - l_{i,t-1}) + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{i,t-1}^L \right] - \gamma'_L \mathbf{D}_{i,t-1}. \end{aligned} \quad (24)$$

This equation serves as a proxy for the innovation term. We estimate the parameter vector $\left(\frac{\beta_L}{\delta}, \frac{\nu}{\delta}, \rho_L, \gamma'_L \right)'$ by nonlinear GMM. The instrument vector is

$$\mathbf{Q}_{L,it} = (1, m_{i,t-1} - l_{i,t-1}, SC_{i,t-1}^L, \mathbf{D}'_{i,t-1})', \quad (25)$$

which contains a constant, the lagged input ratio, the lagged labor cost share, and the lagged digitalization variables. The corresponding moment conditions are

$$\mathbb{E} \left[\mathbf{Q}_{L,it} \xi_{L,it} \left(\frac{\beta_L}{\delta}, \frac{\nu}{\delta}, \rho_L, \gamma_L \right) \right] = \mathbf{0}. \quad (26)$$

The number of instruments equals the number of parameters in this block. Step 2 is therefore exactly identified under the maintained exclusion and rank conditions.

Since $\xi_{L,it}$ does not contain Y_{it} , the variables dated $t - 1$ in (25) are valid instruments under the maintained information assumptions. Thus Step 2 provides estimates of β_L/δ and ν/δ , together with ρ_L and γ_L . Step 1 has already identified $\nu = \beta_L + \beta_M$. We can therefore recover the production function effects as

$$\widehat{\delta} = \frac{\widehat{\nu}}{(\widehat{\nu/\delta})}, \quad \widehat{\beta}_L = (\widehat{\beta_L/\delta})\widehat{\delta}, \quad \widehat{\beta}_M = \widehat{\nu} - \widehat{\beta}_L. \quad (27)$$

Thus, the first two steps identify δ , β_L , and β_M , as well as the persistence and digitalization parameters in the law of motion for labor-augmenting productivity.

4.3.3 Step 3: Hicks-neutral productivity

Hicks-neutral productivity dynamics follows

$$\omega_{N,it} = \rho_{0N} + \rho_N \omega_{N,i,t-1} + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it}, \quad \mathbb{E}[\xi_{N,it} | \mathcal{I}_{i,t-1}] = 0. \quad (28)$$

Two representations of $\omega_{N,it}$ are available. From the production function,

$$y_{it}^\dagger \equiv y_{it} - \beta_M m_{it} - \beta_L(l_{it} + \omega_{L,it}) + \frac{1}{2}\delta(m_{it} - l_{it} - \omega_{L,it})^2 = \alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \omega_{N,it} + \eta_{it}. \quad (29)$$

Using (29) we express

$$\omega_{N,it} = y_{it}^\dagger - [\alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \eta_{it}] \quad (30)$$

Alternatively, we can use materials demand to proxy for unobserved Hicks-neutral productivity. Solving (11) for $\omega_{N,it}$ gives the materials-demand inversion

$$\begin{aligned} m_{it}^\dagger &\equiv (p_{it}^M - p_{it}^Y) - \ln \theta - \ln[\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})] + (1 - \beta_M) m_{it} \\ &\quad - \beta_L(l_{it} + \omega_{L,it}) + \frac{1}{2}\delta(m_{it} - l_{it} - \omega_{L,it})^2 = \alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \omega_{N,it}, \end{aligned} \quad (31)$$

where p_{it}^M and p_{it}^Y are log input and output prices. We can use (31) to express $\omega_{N,it}$ which is

$$\omega_{N,it} = m_{it}^\dagger - [\alpha_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2]. \quad (32)$$

The inversion (32) does not contain Y_{it} and is therefore free of η_{it} , which is precisely why it, rather than (30), is used to replace lagged Hicks-neutral productivity. Substituting $\omega_{N,i,t-1} = m_{i,t-1}^\dagger - \alpha_0 - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2$ into (28) and then into (29) gives

$$\begin{aligned} y_{it}^\dagger &= \pi_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \rho_N \left[m_{i,t-1}^\dagger - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2 \right] \\ &\quad + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it} + \eta_{it}, \end{aligned} \quad (33)$$

where $\pi_0 = (1 - \rho_N)\alpha_0$. This specification appears in the Doraszelski and Jaumandreu (2019), Malikov et al. (2024). We estimate this equation by profiled nonlinear least squares in $(\beta_K, \beta_{KK}, \pi_0, \rho_N, \gamma_N)$, holding the step-1 and step-2 estimates fixed. Because the composite error is $\xi_{N,it} + \eta_{it}$ and k_{it} is predetermined, no additional instruments are required. In particular, competitive Step 3 has no $\mathbf{Q}_{N,it}$ instrument vector: it is profiled nonlinear least squares, not an IV or GMM. Standard errors for the full sequence are obtained by a firm-level block bootstrap in which all three steps are re-estimated.

The two productivity series are recovered observation by observation as

$$\widehat{\omega}_{L,it} = m_{it} - l_{it} + \frac{\widehat{\beta}_L}{\widehat{\delta}} - \frac{\widehat{\nu}}{\widehat{\delta}} SC_{it}^L, \quad (34)$$

$$\widehat{\omega}_{N,it} = \widehat{m}_{it}^\dagger - [\widehat{\alpha}_0 + \widehat{\beta}_K k_{it} + \frac{1}{2} \widehat{\beta}_{KK} k_{it}^2] \quad (35)$$

4.4 IPC model

The IPC extension retains exactly the same output-shock structure as the PC benchmark. The firm chooses labor and materials before η_{it} is realized, using the expected profit maximization behavior under the conditional expectation $\theta = \mathbb{E}[e^{\eta_{it}} \mid \mathcal{I}_{it}]$.

This is where our identification strategy departs from the existing IPC formulation. The revenue-share system alone cannot separately identify the technology parameter ν and the firm-level markup μ_{it} . Rather than identifying them jointly from this system, we first identify ν from the quantity side of the model and then use the revenue-share equation to construct the firm-year markup signal. Section 5.5 provides the full identification argument.

4.4.1 FOCs and markup identification

To model markup, we assume that firm i faces the inverse demand function $P_{it} = P(Y_{it}^*, z_{it})$, where z_{it} denotes demand shifters. Because the multiplicative shock is realized after variable inputs are chosen, expected-profit maximization introduces the same factor θ as in the PC benchmark. Writing MC_{it} for marginal cost and markup $\mu_{it} = P_{it}/MC_{it}$, the variable-input first-order conditions of expected profit maximization can be written as

$$\theta MC_{it} \frac{\partial Y_{it}^*}{\partial j_{it}} = P_{it}^j, \quad j \in \{L, M\}. \quad (36)$$

Multiplying equation (36) by $j_{it}/(P_{it} Y_{it}^*)$ gives

$$\theta \frac{MC_{it}}{P_{it}} \left(\frac{\partial Y_{it}^*}{\partial j_{it}} \frac{j_{it}}{Y_{it}^*} \right) = \frac{P_{it}^j j_{it}}{P_{it} Y_{it}^*}. \quad (37)$$

Thus, the observed revenue shares are

$$s_{it}^j \equiv \frac{P_{it}^j j_{it}}{P_{it} Y_{it}^*} = \theta \frac{\varepsilon_{it}^j}{\mu_{it}} e^{-\eta_{it}}, \quad j \in \{L, M\}. \quad (38)$$

consequently, either variable input provides a noisy signal of the same markup ($\mu_{it} e^{\eta_{it}}$):

$$\frac{\varepsilon_{it}^L}{s_{it}^L} = \frac{\varepsilon_{it}^M}{s_{it}^M} = \frac{\mu_{it}}{\theta} e^{\eta_{it}}. \quad (39)$$

The ratio of the labor and materials conditions removes μ_{it} and η_{it} :

$$\frac{s_{it}^L}{s_{it}^M} = \frac{\varepsilon_{it}^L}{\varepsilon_{it}^M} = \frac{\beta_L + \delta(m_{it} - l_{it} - \omega_{L,it})}{\beta_M - \delta(m_{it} - l_{it} - \omega_{L,it})}. \quad (40)$$

This is the same as (14). Hence the labor-augmenting productivity inversion is unchanged. Equation (40) gives

$$\omega_{L,it} = m_{it} - l_{it} + \frac{\beta_L}{\delta} - \frac{\nu}{\delta} SC_{it}^L. \quad (41)$$

This is identical to (16). As before, using $\nu = \beta_L + \beta_M$ gives

$$\varepsilon_{it}^L = \nu SC_{it}^L, \quad \varepsilon_{it}^M = \nu SC_{it}^M. \quad (42)$$

Adding the two observed shares gives

$$R_{it} \equiv s_{it}^L + s_{it}^M = \theta \frac{\nu}{\mu_{it}} e^{-\eta_{it}}. \quad (43)$$

Therefore,

$$RR_{it} \equiv \ln(\theta\nu) - \ln R_{it} = \ln \mu_{it} + \eta_{it}. \quad (44)$$

which is different from (12) because of the presence of markup μ_{it} .

5 Identification and Estimation

The IPC estimator proceeds in four stages. Hicks-neutral technology and dynamics, labor-augmenting dynamics conditional on that technology, markup dynamics, and signal extraction. The PC steps in Section 4.3 describe a separate benchmark.

5.1 Stage 1: Hicks-neutral technology and dynamics

Combining equation (19) with $y_{it} = y_{it}^* + \eta_{it}$ gives

$$y_{it} = \kappa_0 + \beta_K k_{it} + \frac{1}{2} \beta_{KK} k_{it}^2 + \nu m_{it} - \left(\frac{\nu^2}{2\delta} \right) (SC_{it}^L)^2 + \omega_{N,it} + \eta_{it}. \quad (45)$$

Hicks-neutral productivity is assumed to follow

$$\omega_{N,it} = \pi_N + \rho_N \omega_{N,i,t-1} + \gamma'_N \mathbf{D}_{i,t-1} + \xi_{N,it}, \quad \mathbb{E}[\xi_{N,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (46)$$

Quasi differencing equation (45) gives

$$\begin{aligned}
u_{N,it} = & y_{it} - \pi_0 - \rho_N y_{i,t-1} - \beta_K(k_{it} - \rho_N k_{i,t-1}) \\
& - \frac{1}{2}\beta_{KK}(k_{it}^2 - \rho_N k_{i,t-1}^2) - \nu(m_{it} - \rho_N m_{i,t-1}) \\
& + \left(\frac{\nu^2}{2\delta}\right) [(SC_{it}^L)^2 - \rho_N (SC_{i,t-1}^L)^2] - \gamma'_N \mathbf{D}_{i,t-1},
\end{aligned} \tag{47}$$

where $\pi_0 = \pi_N + (1 - \rho_N)\kappa_0$ and the composite disturbance is

$$u_{N,it} = \xi_{N,it} + \eta_{it} - \rho_N \eta_{i,t-1}. \tag{48}$$

The Stage 1 parameter vector is

$$\boldsymbol{\theta}_N = (\pi_0, \beta_K, \beta_{KK}, \nu, \delta, \rho_N, \gamma_N)'. \tag{49}$$

We estimate this block by nonlinear GMM using

$$\mathbb{E}[\mathbf{Z}_{N,it} u_{N,it}] = \mathbf{0}, \tag{50}$$

where $\mathbf{Z}_{N,it}$ contains predetermined and sufficiently dated variables that are orthogonal to $\xi_{N,it}$, η_{it} , and $\eta_{i,t-1}$. In particular, the presence of $\eta_{i,t-1}$ in equation (48) requires the timing of the instruments to be chosen accordingly.

The common instrument block uses capital dated $t - 1$ and its square, together with the $t - 2$ labor cost share, digitalization variables, and materials. Under the stated timing assumptions, lagged capital and its square are orthogonal to both η_{it} and $\eta_{i,t-1}$ and to the current Hicks-neutral innovation $\xi_{N,it}$.

$$\mathbf{Z}_{N,it}^0 = \left(1, (SC_{i,t-2}^L)^2, \mathbf{D}_{i,t-2}', m_{i,t-2}, k_{i,t-1}, k_{i,t-1}^2\right)', \tag{51}$$

and $\mathbf{Z}_{N,it}$ stacks $\mathbf{Z}_{N,it}^0$ with the sample-specific instruments in Table 3.

Table 3: Hicks-neutral productivity with sample-specific instruments

Sample	Sample-specific instruments added to $\mathbf{Z}_{N,it}^0$	Moments / free parameters	
		Composite	Four components
Overall	$R_{i,t-2}, ICT_{i,t-2}$	8 / 7	11 / 10
Small firms	$SC_{i,t-2}^L, R_{i,t-2}$	8 / 7	11 / 10
Large firms	$s_{i,t-2}^L, \ln(\text{effective hours}_{i,t-2})$	8 / 7	11 / 10
Low-digital industries	$R_{i,t-2}, ICT_{i,t-2}, \mathbf{1}\{\text{metal products}\}_i$	9 / 7	12 / 10
High-digital industries	$s_{i,t-2}^L, k_{i,t-1} m_{i,t-2}$	8 / 7	11 / 10

Notes: Every sample uses the common instruments $\mathbf{Z}_{N,it}^0$ in (51). $R_{i,t-2} = s_{i,t-2}^L + s_{i,t-2}^M$ is the lagged sum of the labor and materials expenditure shares of revenue; $s_{i,t-2}^L$ is the lagged labor expenditure share of revenue; $ICT_{i,t-2}$ is the lagged ICT index built from the IT-technology, digital-human-capital and stakeholder-facing items of the digitalization index; $\ln(\text{effective hours}_{i,t-2})$ is lagged log total effective hours; $\mathbf{1}\{\text{metal products}\}_i$ indicates the metal and metal products industry within the low-digital sample. In the component specification $\mathbf{D}_{i,t-2}$ contains the four lagged component indices and the sample-specific instruments are unchanged.

After estimating the technology parameters, we construct the firm-year Hicks-neutral productivity signal

$$H_{it} \equiv y_{it} - \left[\hat{\kappa}_0 + \hat{\beta}_K k_{it} + \frac{1}{2} \hat{\beta}_{KK} k_{it}^2 + \hat{\nu} m_{it} - \left(\frac{\hat{\nu}^2}{2\hat{\delta}} \right) (SC_{it}^L)^2 \right]. \quad (52)$$

By construction,

$$H_{it} = \omega_{N,it} + \eta_{it}. \quad (53)$$

Stage 1 identifies technology and constructs a Hicks-neutral productivity signal containing the transitory disturbance. Signal extraction in Stage 4 supplies the reported decomposition into persistent productivity and transitory noise.

5.2 Stage 2: Labor-augmenting dynamics

Stage 2 uses the Stage 1 Hicks-neutral estimates $(\hat{\nu}, \hat{\delta})$ as fixed inputs when estimating labor-augmenting productivity dynamics. Labor-augmenting productivity follows the controlled first-order Markov process

$$\omega_{L,it} = \rho_L \omega_{L,i,t-1} + \gamma'_L \mathbf{D}_{i,t-1} + \xi_{L,it}, \quad \mathbb{E}[\xi_{L,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (54)$$

Evaluated at the Stage 1 estimates of ν and δ , it holds in period t ,

$$\omega_{L,it} = (m_{it} - l_{it}) + \frac{\beta_L}{\hat{\delta}} - \frac{\hat{\nu}}{\hat{\delta}} SC_{it}^L, \quad (55)$$

$$\begin{aligned} (m_{it} - l_{it}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{it}^L = & - (1 - \rho_L) \frac{\beta_L}{\hat{\delta}} + \rho_L \left[(m_{i,t-1} - l_{i,t-1}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{i,t-1}^L \right] \\ & + \gamma'_L \mathbf{D}_{i,t-1} + \xi_{L,it}. \end{aligned} \quad (56)$$

Solving (56) for the innovation gives the residual

$$\begin{aligned} \xi_{L,it} = & (m_{it} - l_{it}) - \rho_L (m_{i,t-1} - l_{i,t-1}) + (1 - \rho_L) \frac{\beta_L}{\hat{\delta}} \\ & - \frac{\hat{\nu}}{\hat{\delta}} (SC_{it}^L - \rho_L SC_{i,t-1}^L) - \gamma'_L \mathbf{D}_{i,t-1}, \end{aligned} \quad (57)$$

which depends on the parameter vector

$$\boldsymbol{\theta}_L = (\beta_L, \rho_L, \gamma'_L)'. \quad (58)$$

We use the instrument vector

$$\mathbf{Z}_{L,it} = \left(1, (m_{i,t-1} - l_{i,t-1}) - \frac{\hat{\nu}}{\hat{\delta}} SC_{i,t-1}^L, \mathbf{D}'_{i,t-1} \right)', \quad (59)$$

with moment conditions

$$\mathbb{E}[\mathbf{Z}_{L,it}\xi_{L,it}] = \mathbf{0}. \quad (60)$$

5.3 Stage 3: Markup dynamics and the second signal

We allow the log markup to follow

$$\ln \mu_{it} = \pi_\mu + \rho_\mu \ln \mu_{i,t-1} + \gamma_\mu D_{i,t-1} + \xi_{\mu,it}, \quad \mathbb{E}[\xi_{\mu,it} \mid \mathcal{I}_{i,t-1}] = 0. \quad (61)$$

Using the Stage 1 estimate of ν , equation (44) implies the markup signal $RR_{it} = \ln(\theta\nu) - \ln R_{it} = \ln \mu_{it} + \eta_{it}$. Because θ is constant, the empirically unadjusted signal is

$$\widetilde{RR}_{it} \equiv \ln \nu - \ln R_{it} = \ln \mu_{it} + \eta_{it} - \ln \theta \quad (62)$$

Stage 3 estimates the markup transition by linear instrumental variables. Write $\widetilde{\pi}_\mu = \pi_\mu - (1 - \rho_\mu) \ln \theta$. The observed-signal residual is

$$u_{\mu,it} = \widetilde{RR}_{it} - \widetilde{\pi}_\mu - \rho_\mu \widetilde{RR}_{i,t-1} - \gamma_\mu D_{i,t-1} = \xi_{\mu,it} + \eta_{it} - \rho_\mu \eta_{i,t-1}. \quad (63)$$

The lagged observed signal $\widetilde{RR}_{i,t-1}$ contains $\eta_{i,t-1}$. The same disturbance enters $u_{\mu,it}$ with coefficient $-\rho_\mu$. The lagged signal is therefore endogenous in the transition equation. We use

$$\mathbf{Z}_{\mu,it} = \left(1, \widetilde{RR}_{i,t-2}, D_{i,t-2}\right)', \quad \mathbb{E}[\mathbf{Z}_{\mu,it}u_{\mu,it}] = \mathbf{0}. \quad (64)$$

These moments require the instruments to be orthogonal to both the current markup innovation and the current and lagged transitory disturbances. The four-component specification replaces D with the vector of component indices. Both specifications are exactly identified. Each firm-block bootstrap replication repeats the preceding stages and this markup estimation.

The Hicks-neutral and markup blocks therefore provide two observed signals with the same transitory output shock:

$$H_{it} = \omega_{N,it} + \eta_{it}, \quad \widetilde{RR}_{it} = (\ln \mu_{it} - \ln \theta) + \eta_{it}. \quad (65)$$

Their difference removes the common error,

$$H_{it} - \widetilde{RR}_{it} = \omega_{N,it} - \ln \mu_{it} + \ln \theta. \quad (66)$$

This relation provides additional information for separating Hicks-neutral productivity from the markup.

5.4 Stage 4: The state-space decomposition

The baseline implementation uses the state-space step to produce the firm-year Hicks-neutral productivity

and markup summaries in Table 10, Figure 1, and Appendix A. The structural technology and transition estimates are obtained before smoothing. The decomposition requires the additional second-moment restrictions in Appendix C. Define the state vector

$$\mathbf{x}_{it} = \begin{pmatrix} \omega_{N,it} \\ \widetilde{m}_{it}^\mu \end{pmatrix}. \quad (67)$$

The state equation is

$$\mathbf{x}_{it} = \begin{pmatrix} \pi_N \\ \widetilde{\pi}_\mu \end{pmatrix} + \begin{pmatrix} \rho_N & 0 \\ 0 & \rho_\mu \end{pmatrix} \mathbf{x}_{i,t-1} + \begin{pmatrix} \gamma_N \\ \gamma_\mu \end{pmatrix} D_{i,t-1} + \begin{pmatrix} \xi_{N,it} \\ \xi_{\mu,it} \end{pmatrix}. \quad (68)$$

The measurement system is

$$\begin{pmatrix} H_{it} \\ \widetilde{RR}_{it} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x}_{it} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \eta_{it}. \quad (69)$$

We assume η_{it} has mean zero and is serially independent. It is also orthogonal to the state innovations $\xi_{N,it}$ and $\xi_{\mu,it}$. The state innovations have mean zero and a finite covariance matrix. Under Assumption C.2, the transition dynamics and lag-one residual covariances identify the disturbance variances. At least one nonzero persistence parameter is sufficient to recover the common transitory variance. Using all own- and cross-covariance routes requires both persistence parameters to be nonzero. Proposition C.2 derives these population restrictions.

Under those additional restrictions, the Kalman filter and smoother provide minimum-mean-squared-error linear predictions:

$$\widehat{\omega}_{N,it} = \text{Proj}_{\text{lin}}(\omega_{N,it} \mid \mathcal{Y}_i), \quad \widehat{\widetilde{m}}_{it}^\mu = \text{Proj}_{\text{lin}}(\widetilde{m}_{it}^\mu \mid \mathcal{Y}_i). \quad (70)$$

Here $\mathcal{Y}_i$ contains the two signal sequences for firm i , and Proj_{lin} denotes their linear projection. The common transitory disturbance is predicted from either measurement equation:

$$\widehat{\eta}_{it} = H_{it} - \widehat{\omega}_{N,it} = \widetilde{RR}_{it} - \widehat{\widetilde{m}}_{it}^\mu. \quad (71)$$

Labor-augmenting productivity does not require the state-space decomposition because both markup and the transitory output shock cancel from the ratio of the two variable input conditions. Its firm-year prediction uses the Stage 1 Hicks-neutral estimates $\widehat{\nu}$ and $\widehat{\delta}$:

$$\widehat{\omega}_{L,it} = m_{it} - l_{it} + \frac{\widehat{\beta}_L}{\widehat{\delta}} - \frac{\widehat{\nu}}{\widehat{\delta}} SC_{it}^L. \quad (72)$$

Thus, the IPC model identifies labor-augmenting productivity directly and identifies noisy firm-year signals for Hicks-neutral productivity and the log markup. Separate firm-year predictions of Hicks-neutral

productivity, the markup, and transitory output shock require the additional state-space restrictions in Appendix C.

Relation to existing approaches. Our identification strategy is related to several approaches in the production-function literature. Like Doraszelski and Jaumandreu (2019), we allow labor-augmenting and Hicks-neutral productivity.

Our identification builds on the system approach of Malikov et al. (2024), but addresses the additional identification problem created by imperfect competition. In particular, the revenue-share equation alone cannot separately identify the technology parameter ν and the firm-level markup.

Our approach separates these two objects sequentially and then uses their dynamic structure for signal extraction. After identifying the technology block from the quantity side, the revenue-share and production equations provide noisy signals of markup and Hicks-neutral productivity. A state-space representation, together with Kalman filtering and smoothing, separates the persistent latent states from the common transitory shock and yields observation-specific estimates of markup and Hicks-neutral productivity. This signal-extraction step provides information that the structural GMM equations alone cannot recover.

Other approaches obtain identification under market power by imposing additional structure. Demirer (2025) uses a control-function approach to study non-neutral productivity under IPC. Our approach instead retains the proxy-variable structure. Flynn et al. (2019) obtain identification under market power by imposing constant returns to scale. We do not impose this restriction. Returns to scale are estimated along with the other technology parameters. This distinction is relevant for the empirical results reported in the next section.

5.5 Identification Contribution and Relation to Existing Literature

This subsection relates our identification strategy to the existing proxy-variable literature on non-neutral productivity. The key distinction arises under imperfect competition. Under perfect competition, the revenue-share system can identify $\nu = \beta_L + \beta_M$. Under IPC, however, the corresponding equation identifies only ν/μ_{it} , so ν and the firm-level markup cannot be separately identified from that equation alone.

Malikov et al. (2024) address this problem through set identification and also discuss an alternative price-based materials-demand approach. The former provides bounds on the technology parameters, while the latter requires firm-level variation in relative input prices. Our approach instead obtains point identification of ν from the quantity side of the model. Given ν , the revenue-share equation can then be used to recover the markup signal.

We do not use the revenue-share equation to identify ν . Instead, ν (together with β_K , β_{KK} , and the Hicks-neutral transition parameters) is identified in Section 5.1 from the *quantity side* of the model, i.e., the dynamic production equation (45) combined with the Markovian law of motion for $\omega_{N,it}$, a block that does not involve the markup. Once we obtain ν , we use the revenue-share identity in equation (44) to recover a signal of $\ln \mu_{it}$. Sequential identification also provides the signals used in Stage 4. The production and

revenue-share equations yield Hicks-neutral and markup signals that share a transitory disturbance under the maintained model. Proposition C.2 derives the implied own- and cross-covariance restrictions.

To summarize, our identification strategy has two distinct layers. First, we identify ν from the quantity side of the model and only then use the revenue-share equation to construct the markup signal. This sequencing separates technology identification from markup identification. Second, the state-space model exploits the joint dynamics of the markup and Hicks-neutral-productivity signals. Under the additional second-moment restrictions, Kalman filtering and smoothing separate the common transitory shock and deliver observation-specific estimates of $\ln \mu_{it}$ and $\omega_{N,it}$. The combination of these two steps, rather than a modification of a single revenue-share equation, is the paper’s main identification contribution.

6 Empirical Results

We first report the results under perfect competition (PC). We then turn to imperfect competition (IPC), where we separate productivity from markup dynamics. For both models, we first use the overall digitalization index. We then examine differences across firms and the four components of digitalization.

6.1 Perfect-competition (PC)

6.1.1 Digitalization and productivity

Table 4 reports the effects of digitalization in the PC productivity transition processes, conditional on current productivity. A one-unit increase covers the index’s entire zero-to-one range. A 0.10-point increase corresponds to one tenth of the reported effect. The overall labor-augmenting estimate of 0.037 therefore implies approximately 0.37% higher next-period labor-augmenting productivity. Persistence allows the effect of a sustained increase in adoption to accumulate over time.

Under PC, the overall labor-augmenting estimate is 0.037 with a bootstrap standard error of 0.007. The overall Hicks-neutral estimate is 0.062 with a bootstrap standard error of 0.008. The labor-augmenting estimates are 0.046 for small firms and 0.064 for high-digital industries. These estimates are significant. The estimates for large firms and low-digital industries are not distinguishable from zero. Hicks-neutral estimates are positive and significant in every subgroup.

Table 4: Digitalization effects in productivity transitions under PC

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.037*** (0.007)	0.046*** (0.009)	−0.010 (0.015)	0.016 (0.011)	0.064*** (0.012)
Hicks-neutral productivity	0.062*** (0.008)	0.067*** (0.011)	0.033** (0.014)	0.061*** (0.011)	0.062*** (0.014)
Observations	17,087	12,246	4,343	10,610	5,883
Firms	2,213	1,689	546	1,374	789

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. The index ranges from zero to one. A 0.10-point increase corresponds to one tenth of the reported effect. PC uses one lag and the minimal exactly identified MZZ labor instrument set.

6.1.2 Heterogeneity in the effects of digitalization in the PC

Table 5 tests whether the digitalization effects reported in Table 4 differ across firm-size classifications and between low- and high-digital-intensity industries. Because productivity is estimated allowing for different technological parameters across samples, baseline productivity levels are not directly comparable across subgroups. Consequently, our comparisons focus strictly on the marginal effects of digitalization.

The evidence of heterogeneity is strongest for labor-augmenting productivity. Under PC, the digitalization effect is 0.056 larger for small firms than for large firms, a difference significant at the one percent level. The small-firm effect is positive and significant, while the large-firm effect is not statistically distinguishable from zero. Across industries, the labor-augmenting effect is 0.048 larger in high-digital-intensity industries than in low-digital-intensity industries.

The PC Hicks-neutral effect is 0.034 larger for small firms than for large firms. The bootstrap standard error of this difference is 0.017. The high-minus-low industry difference is 0.001 with a bootstrap standard error of 0.017.

Table 5: Differences in digitalization effects under PC

Outcome	Small	Large	Small–large	High	Low	High–low
Labor-augmenting productivity	0.046*** (0.009)	−0.010 (0.015)	0.056*** (0.017)	0.064*** (0.012)	0.016 (0.011)	0.048*** (0.016)
Hicks-neutral productivity	0.067*** (0.011)	0.033** (0.014)	0.034** (0.017)	0.062*** (0.014)	0.061*** (0.011)	0.001 (0.017)

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. Differences use a joint union-firm bootstrap, preserving covariance when firms appear in both groups. Industry digital intensity does not classify individual firms by their own adoption scores. All differences use 499 bootstrap replications.

6.1.3 Effects of Digitalization Dimensions on Productivity

We next replace the composite index with its four components, described in Section 3, which enter both productivity transitions jointly. Table 6 reports the estimates under PC, with the labor-augmenting process in Panel A and the Hicks-neutral process in Panel B. Because the components enter together, each effect measures the estimated effect of one dimension of digitalization on productivity, holding the other three constant.

PC labor-augmenting effects vary across components. The estimated effect of IT technologies is 0.010 overall. The estimated effect of IT technologies is 0.019 in high-digital industries. Estimates for digital human capital are 0.019 overall and 0.021 for small firms. They are 0.021 in low-digital industries and 0.020 in high-digital industries. Estimates for automation are 0.010 overall and 0.018 in high-digital industries. The estimated effect of automation is 0.019 for small firms. The stakeholder-facing digitalization estimates cannot be distinguished from zero in any sample.

PC Hicks-neutral effects also vary across components. Estimates for IT technologies are 0.011 overall and 0.017 for large firms. The estimated effect of IT technologies is 0.023 in high-digital industries. The estimated effect of digital human capital is 0.019 in high-digital industries. Estimates for automation are

0.018 overall and 0.022 for small firms. The effect is 0.027 in low-digital industries. Estimates for stakeholder-facing digitalization are 0.021 overall and 0.026 for small firms. The estimated effect of stakeholder-facing digitalization is 0.017 in low-digital industries. The estimated effect of stakeholder-facing digitalization is 0.028 in high-digital industries.

The PC component estimates show distinct productivity channels. Digital human capital contributes to labor-augmenting productivity in most samples, while IT technologies and automation are significant in selected samples. Automation and stakeholder-facing digitalization have significant Hicks-neutral effects in the overall and small-firm samples. In high-digital industries, IT technologies and stakeholder-facing digitalization are significant at five percent or better, whereas digital human capital is significant only at ten percent. These results do not attribute labor-augmenting gains exclusively to digital human capital.

Table 6: Digitalization components and productivity effects under PC

Component	Overall	Small	Large	Low	High
<i>Labor-augmenting productivity</i>					
IT technologies	0.010** (0.005)	0.008 (0.006)	0.002 (0.010)	0.006 (0.006)	0.019*** (0.007)
Digital human capital	0.019*** (0.004)	0.021*** (0.006)	0.008 (0.007)	0.021*** (0.005)	0.020*** (0.007)
Automation	0.010** (0.004)	0.019*** (0.005)	-0.007 (0.007)	0.005 (0.006)	0.018** (0.007)
Stakeholder-facing digitalization	0.007 (0.007)	0.009 (0.008)	0.000 (0.011)	-0.007 (0.009)	0.016 (0.010)
<i>Hicks-neutral productivity</i>					
IT technologies	0.011* (0.005)	0.007 (0.007)	0.017* (0.009)	0.005 (0.007)	0.023*** (0.009)
Digital human capital	0.009 (0.006)	0.008 (0.008)	0.010 (0.009)	0.003 (0.007)	0.019* (0.010)
Automation	0.018*** (0.005)	0.022*** (0.006)	0.002 (0.009)	0.027*** (0.007)	0.000 (0.008)
Stakeholder-facing digitalization	0.021*** (0.007)	0.026*** (0.009)	0.005 (0.012)	0.017* (0.009)	0.028** (0.011)
Observations	15,874	11,389	3,988	9,871	5,459
Firms	2,140	1,625	525	1,328	762

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices.

6.2 IPC

In this section, we present the results under IPC.

6.2.1 Digitalization and productivity (IPC)

Table 7 presents the IPC labor-augmenting productivity estimates across the full sample and key structural subsamples. In the pooled baseline specification, the labor-augmenting coefficient is 0.055 with a bootstrap standard error of 0.007. Disaggregating the sample by firm scale reveals substantial heterogeneity: the positive effect is driven entirely by small firms, where the estimate rises to 0.059 with a standard error of 0.008. In contrast, the effect disappears among large firms, yielding a coefficient of -0.002 with a standard

error of 0.011, rendering it statistically indistinguishable from zero. Across sector-level digital intensity, labor-augmenting returns remain statistically significant in both regimes, measuring 0.050 with a standard error of 0.010 in low-digital industries and rising modestly to 0.062 with a standard error of 0.017 in high-digital environments.

For the overall sample, the Hicks-neutral estimate is 0.043 with a bootstrap standard error of 0.008. For small firms, the Hicks-neutral estimate is 0.042 with a bootstrap standard error of 0.010. For large firms, the Hicks-neutral estimate is 0.054 with a bootstrap standard error of 0.026. For low-digital industries, the Hicks-neutral estimate is 0.049 with a bootstrap standard error of 0.010. For high-digital industries, the Hicks-neutral estimate is 0.019 with a bootstrap standard error of 0.017. The estimate for high-digital industries is not distinguishable from zero.

Table 7: Digitalization effects in productivity and markup transitions under IPC

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.055*** (0.007)	0.059*** (0.008)	-0.002 (0.011)	0.050*** (0.010)	0.062*** (0.017)
Hicks-neutral productivity	0.043*** (0.008)	0.042*** (0.010)	0.054** (0.026)	0.049*** (0.010)	0.019 (0.017)
Log-markup transition	0.007** (0.003)	0.013** (0.005)	0.003 (0.007)	0.012*** (0.004)	-0.007 (0.004)
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. The index ranges from zero to one. A 0.10-point increase corresponds to one tenth of the reported effect. IPC uses two lags and estimates Hicks-neutral technology first, labor dynamics conditional on that technology second, and markup dynamics third. Bootstrap replications used for productivity, in column order: 499, 499, 499, 499, 499. Bootstrap replications used for markup: 499, 499, 497, 499, 499.

6.2.2 Digitalization and markup dynamics

Table 7 also reports markup-transition effects. For the overall sample, the markup-transition estimate is 0.007 with a bootstrap standard error of 0.003. For small firms, the markup-transition estimate is 0.013 with a bootstrap standard error of 0.005. For large firms, the markup-transition estimate is 0.003 with a bootstrap standard error of 0.007. For low-digital industries, the markup-transition estimate is 0.012 with a bootstrap standard error of 0.004. For high-digital industries, the markup-transition estimate is -0.007 with a bootstrap standard error of 0.004. The markup responses are distinguishable from zero in the overall, small-firm, and low-digital samples. The large-firm and high-digital responses are imprecise.

The estimated markup effect is more positive in low-digital than in high-digital industries. The subgroup test below confirms a significant difference between these effects, but the negative high-digital estimate is not individually significant. The results therefore support a difference in markup effects across industry groups without establishing markup compression in the high-digital group or identifying a particular competition mechanism.

The economic magnitudes are modest. A 0.10-point increase in lagged digitalization has an estimated effect of approximately 0.069% on next-period markup in the overall IPC model, conditional on lagged

markup. This is smaller than the corresponding overall productivity effects. The unconditional annual mean trend in Figure 1 is a different object and also reflects other shocks and changes in sample composition.

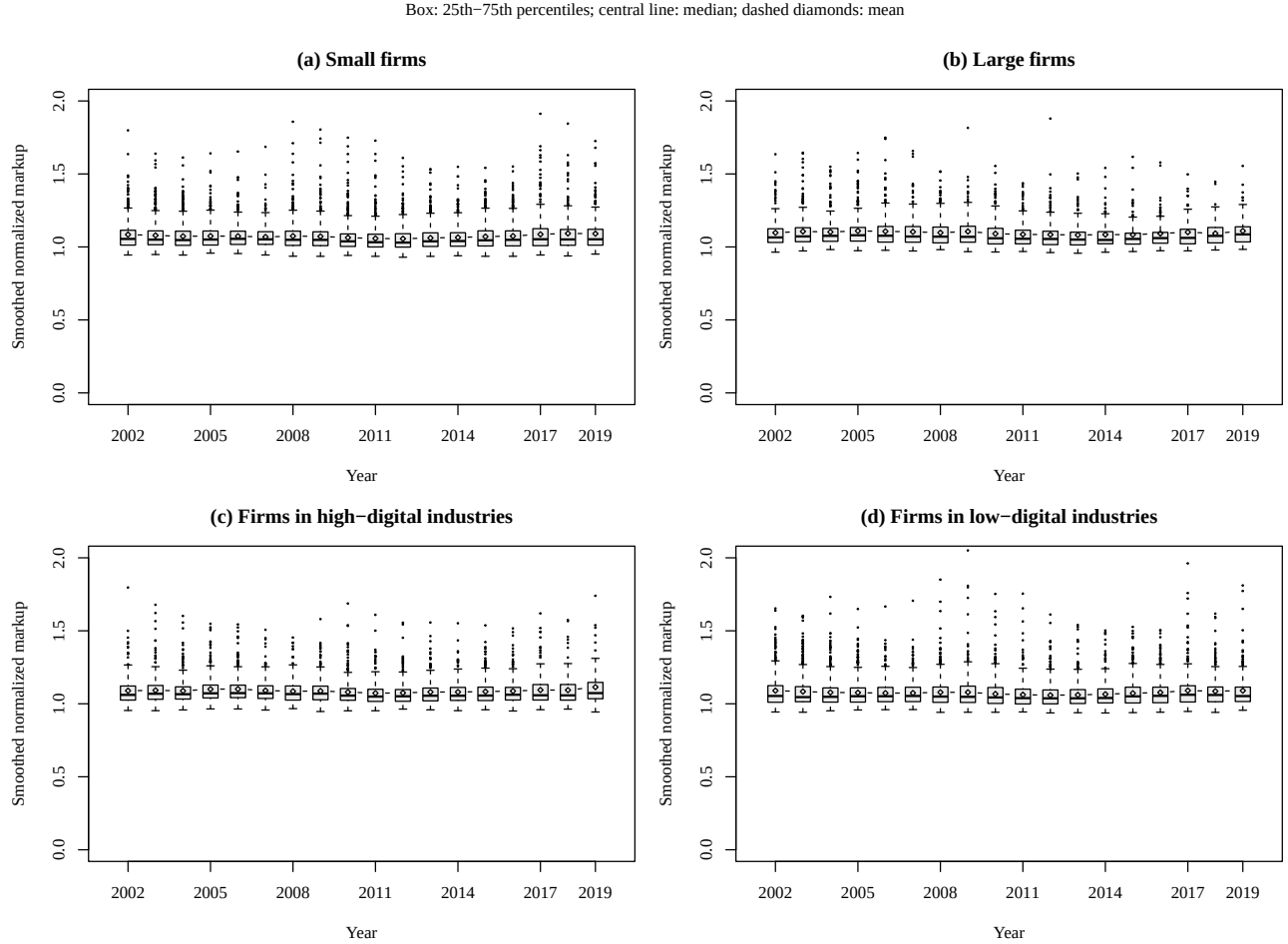


Figure 1: Annual markup distributions by firm size and industry digital intensity

Figure 1 shows annual distributions and mean smoothed normalized markups. For small firms, the annual mean moves from 1.085 in 2002 to its sample-period minimum of 1.056 in 2012. It reaches 1.091 in 2019. The endpoint change is 0.6%. For large firms, the annual mean moves from 1.097 in 2002 to its sample-period minimum of 1.082 in 2013. It reaches 1.110 in 2019. The endpoint change is 1.1%. Technology and markup scale differ across groups. The plotted levels therefore do not rank absolute markups.

For high-digital industries, the annual mean moves from 1.089 in 2002 to its sample-period minimum of 1.072 in 2011. It reaches 1.114 in 2019. The endpoint change is 2.3%. For low-digital industries, the annual mean moves from 1.090 in 2002 to its sample-period minimum of 1.059 in 2012. It reaches 1.090 in 2019. The endpoint change is 0.0%. The groups classify firms by industry digital intensity. Sample composition and technology estimates also affect the comparison.

Calligaris et al. (2018) report rising average markups across 26 countries in 2001–2014 and a widening gap in favor of digitally intensive sectors. Over our overlapping 2002–2014 period, the high-digital mean moves from 1.089 to 1.081, and the low-digital mean from 1.090 to 1.066. The point-normalized series and

changing samples describe within-group evolution; differences in period, industry classification, identification, and normalization preclude a formal comparison of absolute markup levels with their cross-country results.

6.2.3 Heterogeneity in the effects of digitalization under IPC

Table 8 reports the corresponding subgroup comparisons for the IPC effects in Table 7. Under IPC, the small-large labor-augmenting difference is 0.061. The small-firm effect is positive and significant, while the large-firm effect is not statistically distinguishable from zero. The high-low industry difference is 0.012 and is not distinguishable from zero. The estimates therefore provide strong evidence of a firm-size difference, while the industry contrast remains imprecise under IPC. The results are different for Hicks-neutral productivity. The small-large difference is -0.012 , while the high-low industry difference is -0.030 . Neither difference is significant at ten percent. The small-, large-, and low-digital effects are individually significant, while the high-digital effect is not. However, differences in individual significance do not imply significant differences between groups.

The IPC high-minus-low industry markup difference is -0.018 . Its bootstrap standard error is 0.006. The paired test gives $p = 0.002$. The small-minus-large firm difference is 0.010 with a bootstrap standard error of 0.008. It is not distinguishable from zero, with $p = 0.231$. These tests compare groups. They do not establish a markup reduction within either group.

Table 8: Differences in digitalization effects under IPC

Outcome	Small	Large	Small-large	High	Low	High-low
Labor-augmenting productivity	0.059*** (0.008)	-0.002 (0.011)	0.061*** (0.015)	0.062*** (0.017)	0.050*** (0.010)	0.012 (0.020)
Hicks-neutral productivity	0.042*** (0.010)	0.054** (0.026)	-0.012 (0.028)	0.019 (0.017)	0.049*** (0.010)	-0.030 (0.020)
Log markup	0.013** (0.005)	0.003 (0.007)	0.010 (0.008)	-0.007 (0.004)	0.012*** (0.004)	-0.018 *** (0.006)

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. Differences use a joint union-firm bootstrap, preserving covariance when firms appear in both groups. Industry digital intensity does not classify individual firms by their own adoption scores. Bootstrap replications used for markup differences: 499 for size and 499 for industry.

6.2.4 Effects of Digitalization Dimensions on Productivity and Markups

Table 9 reports IPC component estimates for labor-augmenting productivity. Estimates for IT technologies are 0.018 overall and 0.016 for small firms. Estimates for IT technologies are 0.013 in low-digital industries and 0.021 in high-digital industries. Estimates for digital human capital are 0.022 overall and 0.025 for small firms. They are 0.026 in low-digital industries and 0.024 in high-digital industries. Estimates for automation are 0.011 overall and 0.019 for small firms. The estimated effect of automation is 0.015 in high-digital industries. The estimated effect of stakeholder-facing digitalization is 0.011 overall.

The IPC Hicks-neutral results differ across components. The IT technologies estimates cannot be

distinguished from zero in any sample. Estimates for digital human capital are 0.009 overall and 0.010 in low-digital industries. The estimated effect of digital human capital is 0.011 for small firms. Estimates for automation are 0.017 overall and 0.017 for small firms. The effect is 0.022 in low-digital industries. The estimated effect of stakeholder-facing digitalization is 0.013 for small firms.

Automation is the only component with evidence of a markup response. Estimates are 0.009 overall, 0.012 for small firms, and 0.013 in low-digital industries. The other three components cannot be distinguished from zero.

Overall, the IPC component estimates distinguish labor-augmenting, Hicks-neutral, and markup channels. The preceding paragraphs describe component effects by sample, and Table 9 reports all estimates and their bootstrap precision. These results condition on the lagged-capital instrument specification; differences in individual significance do not establish differences between groups.

Table 9: Digitalization components and their effects on productivity and markups under IPC

Component	Overall	Small	Large	Low	High
<i>Labor-augmenting productivity</i>					
IT technologies	0.018*** (0.004)	0.016*** (0.006)	−0.004 (0.019)	0.013** (0.006)	0.021** (0.009)
Digital human capital	0.022*** (0.004)	0.025*** (0.006)	0.012 (0.008)	0.026*** (0.005)	0.024*** (0.009)
Automation	0.011*** (0.004)	0.019*** (0.005)	−0.004 (0.027)	0.008 (0.006)	0.015* (0.009)
Stakeholder-facing digitalization	0.011** (0.006)	0.009 (0.007)	0.004 (0.034)	0.011 (0.007)	0.004 (0.010)
<i>Hicks-neutral productivity</i>					
IT technologies	0.003 (0.004)	0.002 (0.005)	0.008 (0.017)	0.004 (0.006)	0.005 (0.008)
Digital human capital	0.009** (0.004)	0.011* (0.006)	0.015 (0.014)	0.010** (0.005)	0.000 (0.009)
Automation	0.017*** (0.004)	0.017*** (0.006)	0.016 (0.020)	0.022*** (0.005)	0.007 (0.007)
Stakeholder-facing digitalization	0.010 (0.007)	0.013* (0.008)	0.002 (0.022)	0.008 (0.009)	0.011 (0.011)
<i>Log-markup transition</i>					
IT technologies	−0.003 (0.003)	−0.005 (0.003)	0.009 (0.005)	−0.003 (0.004)	0.000 (0.004)
Digital human capital	0.002 (0.002)	0.005 (0.004)	0.002 (0.004)	0.003 (0.003)	−0.004 (0.003)
Automation	0.009*** (0.003)	0.012*** (0.003)	0.003 (0.005)	0.013*** (0.004)	0.004 (0.005)
Stakeholder-facing digitalization	−0.003 (0.004)	0.000 (0.005)	−0.008 (0.006)	−0.002 (0.005)	−0.005 (0.006)
Observations	12,161	8,487	3,085	7,607	4,126
Firms	1,717	1,250	424	1,068	617

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. Four component indices enter each transition jointly. Bootstrap replications used for productivity, in column order: 499, 499, 499, 499, 499. Bootstrap replications used for markup: 499, 499, 497, 499, 499.

6.2.5 Summary of results

Taken together, digitalization has positive overall labor-augmenting and Hicks-neutral effects under both models, alongside a smaller markup effect under IPC. The overall labor-augmenting estimates are 0.037 under PC and 0.055 under IPC; the corresponding Hicks-neutral estimates are 0.062 and 0.043. A 0.10-point increase in lagged digitalization corresponds to approximately 0.37% and 0.62% higher next-period labor-augmenting and Hicks-neutral productivity under PC, and 0.55% and 0.43% under IPC.

The composite labor-augmenting effect is significant among small firms under both models and insignificant among large firms. The small–large differences are 0.056 under PC and 0.061 under IPC, both significant at one percent. The high–low industry labor difference is significant under PC and insignificant under IPC. The PC small–large Hicks-neutral difference is significant at five percent, while neither IPC Hicks-neutral subgroup difference is significant at ten percent. Composite Hicks-neutral effects remain positive in every sample, although the high-digital IPC estimate is imprecise.

6.2.6 Separating transitory shocks from Hicks-neutral productivity

Table 10 reports the baseline state-space decomposition. The markup-based transitory variance estimate is 0.0048, with a bootstrap standard error of 0.0005. Smoothing changes the Hicks-neutral signal standard deviation from 0.1618 to 0.1419. These standard deviations describe firm-year dispersion; the values beneath them are bootstrap standard errors. Appendix C reports the alternative covariance-based variance estimates and equality checks. Their joint restrictions are rejected in the overall, small-firm, and low-digital samples. The large-firm Hicks-neutral estimate is negative and cannot itself be a disturbance variance. We retain the markup-based smoothing results as conditional decompositions rather than validated recovery of the latent states.

Table 10: Transitory-shock extraction under imperfect competition

Outcome	Overall	Small	Large	Low	High
$\hat{\sigma}_\eta^2$	0.0048 (0.0005)	0.0058 (0.0006)	0.0026 (0.0004)	0.0044 (0.0005)	0.0060 (0.0007)
<i>Firm-year means</i>					
Labor-augmenting productivity	0.3173 (0.0351)	0.2443 (0.0326)	-0.0869 (0.1101)	0.3468 (0.0659)	0.2305 (0.1456)
Hicks-neutral signal	0.0647 (0.0059)	0.0621 (0.0068)	0.0683 (0.0205)	0.0708 (0.0120)	0.0605 (0.0167)
Smoothed Hicks-neutral productivity	0.0643 (0.0059)	0.0619 (0.0068)	0.0677 (0.0205)	0.0704 (0.0120)	0.0599 (0.0166)
Predicted transitory shock	0.0004 (0.0001)	0.0003 (0.0002)	0.0006 (0.0002)	0.0004 (0.0001)	0.0005 (0.0004)
Smoothed normalized log markup	0.0666 (0.0098)	0.0658 (0.0167)	0.0878 (0.0704)	0.0681 (0.0145)	0.0806 (0.0728)
<i>Firm-year standard deviations</i>					
Labor-augmenting productivity	0.4686 (0.0136)	0.4410 (0.0146)	0.3697 (0.0797)	0.5069 (0.0346)	0.3739 (0.0996)
Hicks-neutral signal	0.1618 (0.0033)	0.1635 (0.0051)	0.1551 (0.0791)	0.1648 (0.0348)	0.1535 (0.0631)
Smoothed Hicks-neutral productivity	0.1419 (0.0035)	0.1407 (0.0056)	0.1379 (0.0805)	0.1475 (0.0356)	0.1282 (0.0677)
Predicted transitory shock	0.0519 (0.0044)	0.0601 (0.0058)	0.0348 (0.0049)	0.0491 (0.0044)	0.0673 (0.0073)
Smoothed normalized log markup	0.0862 (0.0037)	0.0845 (0.0051)	0.0946 (0.0079)	0.0864 (0.0034)	0.0808 (0.0042)

Notes: Parentheses report firm-block-bootstrap standard errors based on 499 replications repeating all estimation stages. The reported $\hat{\sigma}_\eta^2$ is estimated from the markup residual's lag-one covariance and is used in both measurement equations. Alternative estimates and their equality tests appear in Table 17.

7 Diagnostics and Robustness

We perform robustness checks in this section.

7.1 Technology and persistence estimates

The diagnostics and robustness checks reported in Tables 11 and 12 assess the fitted technology, instrument restrictions, and precision of the estimates. They help qualify the interpretation of the results rather than independently validating the model assumptions.

7.1.1 Perfect competition

Under PC, median returns to scale range from 0.949 to 0.987 across samples. Mean capital elasticity is positive in every sample, ranging from 0.078 to 0.113; no retained firm-year has a negative estimated capital derivative. The sign of β_K alone does not determine monotonicity because $\varepsilon_{it}^K = \beta_K + \beta_{KK}k_{it}$.

The PC persistence estimates range from 0.796 to 0.903 across the two productivity processes and samples. The estimates indicate persistence in both productivity processes.

Table 11: Technology, persistence and estimation diagnostics under PC

Parameter	Overall	Small	Large	Low	High
ρ_L	0.891*** (0.005)	0.884*** (0.005)	0.903*** (0.011)	0.901*** (0.005)	0.862*** (0.010)
ρ_N	0.840*** (0.011)	0.853*** (0.013)	0.796*** (0.027)	0.837*** (0.014)	0.838*** (0.021)
ν	0.872*** (0.002)	0.871*** (0.002)	0.876*** (0.004)	0.872*** (0.002)	0.873*** (0.003)
δ	-0.129*** (0.003)	-0.139*** (0.004)	-0.111*** (0.009)	-0.123*** (0.004)	-0.137*** (0.006)
β_K	-0.129*** (0.041)	-0.159*** (0.062)	-0.058 (0.194)	-0.141*** (0.053)	-0.072 (0.066)
β_{KK}	0.0156*** (0.0029)	0.0174*** (0.0047)	0.0099 (0.0110)	0.0168*** (0.0038)	0.0102** (0.0045)
Median returns to scale	0.969	0.949	0.987	0.977	0.950
Mean capital elasticity	0.098	0.078	0.113	0.106	0.078
Share with negative capital elasticity	0.000	0.000	0.000	0.000	0.000
Observations	17,087	12,246	4,343	10,610	5,883
Firms	2,213	1,689	546	1,374	789

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. Step 2 is exactly identified; Step 3 is nonlinear least squares. Among solutions with positive variable-input effects, we minimize the Step-3 residual sum of squares. Each bootstrap replication uses the same selection rule.

7.1.2 IPC

The IPC Hansen reference p -values range from 0.123 to 1.000. They do not reject the overidentifying restrictions at conventional levels; these reference tests alone do not establish instrument validity.

Mean IPC returns to scale range from 0.971 to 1.003. The large-firm estimate is close to one and slightly above it, while the other point estimates are below one. Mean capital elasticity is positive in every IPC sample, ranging from 0.031 to 0.042. In the large-firm sample, 10.2% of firm-years nevertheless have negative capital elasticity; the other composite samples have none.

IPC labor-augmenting persistence ranges from 0.906 to 0.942, Hicks-neutral persistence from 0.510 to 0.819, and markup persistence from 0.836 to 0.892. The large-firm Hicks-neutral persistence estimate of 0.510 is lower than those in the other samples.

Table 12: Technology, persistence and estimation diagnostics under IPC

Parameter	Overall	Small	Large	Low	High
ρ_L	0.934*** (0.005)	0.919*** (0.006)	0.940*** (0.014)	0.942*** (0.007)	0.906*** (0.016)
ρ_N	0.777*** (0.029)	0.817*** (0.034)	0.510*** (0.172)	0.788*** (0.033)	0.819*** (0.097)
ρ_μ	0.846*** (0.019)	0.859*** (0.024)	0.846*** (0.038)	0.836*** (0.022)	0.892*** (0.032)
ν	0.939*** (0.009)	0.938*** (0.016)	0.961*** (0.092)	0.941*** (0.014)	0.951*** (0.078)
δ	-0.206*** (0.006)	-0.209*** (0.008)	-0.176*** (0.043)	-0.203** (0.089)	-0.198 (0.217)
β_K	-0.016 (0.030)	-0.022 (0.078)	0.554 (1.640)	-0.021 (0.242)	0.017 (0.231)
β_{KK}	0.0038* (0.0021)	0.0041 (0.0060)	-0.0297 (0.0923)	0.0039 (0.0165)	0.0010 (0.0170)
Mean returns to scale	0.978	0.971	1.003	0.978	0.982
Mean capital elasticity	0.039	0.034	0.042	0.037	0.031
Share with negative capital elasticity	0.000	0.000	0.102	0.000	0.000
Hansen reference p	0.686	0.539	1.000	0.522	0.123
Mean $\hat{\nu}/R$	1.078 (0.011)	1.077 (0.019)	1.100 (0.106)	1.078 (0.016)	1.092 (0.089)
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 Alternative materials-demand proxy

The sensitivity specification replaces the main Hicks-neutral block with a lagged materials-demand proxy adjusted using a forecast of lagged markup. Appendix B gives its construction and additional exclusion restrictions. Table 13 reports estimates for the same IPC samples and capital measure.

Table 13: Sensitivity using a markup-adjusted materials proxy

Outcome	Overall	Small	Large	Low	High
Labor-augmenting productivity	0.061*** (0.010)	0.053*** (0.013)	0.006 (0.014)	0.033*** (0.012)	0.055*** (0.013)
Hicks-neutral productivity	0.039*** (0.013)	0.014 (0.010)	0.005 (0.009)	0.016** (0.008)	0.025 (0.017)
Log-markup transition	0.007** (0.003)	0.013*** (0.005)	0.003 (0.006)	0.012*** (0.004)	-0.007 (0.004)
ν	0.118	0.907	1.004	0.909	0.833
ρ_N	0.987	0.950	0.921	0.947	0.963
Draws at $\nu = 0.05$	168	10	3	0	5
Draws at $\rho_N = 0.995$	2	29	0	0	6
Observations	13,291	9,266	3,415	8,278	4,523
Firms	1,816	1,324	457	1,120	656

Notes: Firm-block-bootstrap standard errors are in parentheses, from 499 replications repeating all estimation stages. Stars use two-sided, unadjusted normal-reference tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference is conditional on the sample construction and instrument choices. This comparison uses the IPC composite-index sample and the alternative estimation order in Appendix B. It imposes $0.05 \leq \nu \leq 3$, $-0.95 \leq \rho_N \leq 0.995$, and capital elasticity at least 10^{-6} on the retained support. Capital curvature is estimated in all five groups. Normal-reference stars are not boundary-calibrated.

The alternative-proxy specification gives the following results. For the overall sample, the labor-augmenting estimate is 0.061. For small firms, the labor-augmenting estimate is 0.053. For large firms, the labor-augmenting estimate is 0.006. For low-digital industries, the labor-augmenting estimate is 0.033. For high-digital industries, the labor-augmenting estimate is 0.055. The estimate for large firms is not distinguishable from zero. For the overall sample, the Hicks-neutral estimate is 0.039. For small firms, the Hicks-neutral estimate is 0.014. For large firms, the Hicks-neutral estimate is 0.005. For low-digital industries, the Hicks-neutral estimate is 0.016. For high-digital industries, the Hicks-neutral estimate is 0.025. The estimates for small firms, large firms and high-digital industries are not distinguishable from zero. For the overall sample, the markup-transition estimate is 0.007. For small firms, the markup-transition estimate is 0.013. For large firms, the markup-transition estimate is 0.003. For low-digital industries, the markup-transition estimate is 0.012. For high-digital industries, the markup-transition estimate is -0.007. The estimates for large firms and high-digital industries are not distinguishable from zero.

The alternative estimates labor dynamics before Hicks-neutral technology, whereas the main IPC model estimates Hicks-neutral technology first. The overall labor responses are 0.061 in the alternative and 0.055 in the main model. The alternative also imposes parameter bounds and positive capital elasticity over the estimation support. This comparison changes the estimation order and restrictions as well as the Hicks-neutral proxy, so it is a sensitivity exercise rather than an isolated proxy substitution.

8 Policy Implications

Our findings are relevant for evaluating policies that support digital adoption. The productivity and markup channels should be considered separately. Under perfect competition, a 0.10-point increase in the digitalization index is associated with about 0.37% higher labor-augmenting productivity and 0.62% higher Hicks-neutral productivity in the following period. Under imperfect competition, the corresponding estimates are 0.55% and 0.43%, together with a smaller markup response of 0.07%. These are conditional relationships. They suggest that digital adoption is associated with better use of labor and higher productive capacity. The markup response captures a different channel and can have different implications for how the gains are distributed.

The firm-size results give a specific reason to consider smaller manufacturers when designing adoption support. The labor-augmenting response is significantly larger among firms with fewer than 200 workers than among larger firms under both competitive specifications. Programs that improve access to digital equipment, implementation assistance, and relevant training could therefore help smaller firms strengthen their labor efficiency. This finding does not imply that support for large firms has no productive value: their Hicks-neutral response remains positive and significant. Instead, it suggests that program objectives should reflect the different productivity channels observed across firm sizes. Whether smaller firms should receive greater public support also depends on adoption costs, financing constraints, and the additional investment generated by that support, which our estimates do not directly identify.

The component estimates further suggest that the composition of adoption should inform policy de-

sign. Among small firms, IT technologies, digital human capital, and automation each have positive and significant labor-augmenting responses under imperfect competition. The digital human-capital result provides a reason to evaluate IT training alongside technology acquisition, while the IT and automation results indicate that labor-efficiency gains are associated with several forms of digital investment. Automation also has positive Hicks-neutral and markup responses in the overall sample and in several subgroups. Consequently, support programs should track which technologies firms adopt and which outcomes improve, rather than treating all digital investments as equivalent. These findings motivate evaluating programs that combine skills development with technology access, although they do not directly establish complementarities or determine the optimal allocation of subsidies.

The markup findings make competition conditions relevant to the assessment of digital adoption policies. The estimated markup response is significantly smaller in high-digital-intensity industries than in low-digital-intensity industries. This distinction complements the higher markup levels in digitally intensive industries documented by Calligaris et al. (2018): industries with higher markups need not exhibit a larger response to additional adoption. Competition assessments should therefore consider both existing markup levels and their evolution as firms digitalize. Where adoption is associated with productivity improvements and higher markups, further evidence on marginal costs and consumer prices is needed to establish how much of the productive gain reaches consumers. The negative but individually insignificant estimate in high-digital-intensity industries does not establish that adoption reduces market power.

Finally, the estimates offer quantitative benchmarks for designing and evaluating adoption programs. A 0.10-point increase in the index provides a common scale for reporting the estimated relationships, but it does not represent a fixed expenditure or a particular technology purchase. Because productivity and markups are persistent, evaluations should also examine whether gains accumulate when higher adoption is sustained. Establishing broader policy benefits requires tracking wages, employment, consumer prices, and implementation costs alongside firm productivity and markups. Labor-augmenting productivity alone does not determine wages, employment, or the labor share, while a higher markup alone does not establish a consumer welfare loss. Our estimates identify conditional relationships within the maintained production framework; causal returns to specific interventions require additional evidence. Their policy value lies in identifying promising outcomes and groups to examine, and in providing benchmarks against which intervention-specific results can be assessed.

9 Conclusion

This paper examines digitalization, labor-augmenting and Hicks-neutral productivity, and firm markups under PC and IPC. The overall productivity estimates are positive and significant in both models. The labor-augmenting response is significantly larger among small firms under both specifications. The PC firm-size Hicks-neutral difference is significant at five percent. The other Hicks-neutral subgroup differences are imprecise. Under IPC, the overall markup response is small, positive, and significant. It is significantly lower in high-digital industries than in low-digital industries. The component results also show different

channels. In particular, automation is associated with Hicks-neutral productivity and markups in the overall, small-firm, and low-digital IPC samples.

The digitalization index combines technologies with different productivity effects. Those effects vary by firm size, industry digital intensity, and technology. Separating productivity from markups requires identifying restrictions. Comparing the two models shows how the results depend on those restrictions.

These findings distinguish productivity effects with different implications for the labor share. A labor-augmenting effect alone does not determine that share, which also depends on substitution between inputs and market power. The distinction is relevant to the discussion of labor-complementing technologies in Acemoglu et al. (2026). Our estimates show positive labor-augmenting effects of digital human capital in several samples, while automation has both labor-augmenting and Hicks-neutral effects in some samples. They therefore do not support treating digital human capital as the exclusive labor-augmenting channel or automation as exclusively Hicks-neutral. The results concern firm productivity and markups and do not directly estimate effects on workers or the returns to particular policy interventions.

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Appendix

A Distributions of Elasticities, Productivities and Markups

Table 14 summarizes the PC estimates. In the overall sample, mean output elasticities are 0.098 for capital, 0.235 for labor, and 0.637 for materials. Labor and materials elasticities sum to the estimated variable-input scale ν by construction. Their equal standard deviations follow from these identities rather than providing independent evidence about technology.

The final two columns of Table 14 report recovered log productivity after subtracting each sample's 2001 median. These are overall firm-year distributions relative to a common within-sample reference point. A negative observation is below that reference median. Labor-augmenting productivity has greater dispersion in its own log units than Hicks-neutral productivity, but these components enter output differently, so their dispersions are not directly comparable output contributions.

Table 14: PC elasticities and normalized productivity distributions

Sample	ε_K	ε_L	ε_M	$\tilde{\omega}_L$	$\tilde{\omega}_N$
Overall	0.098 (0.032) [0.073, 0.097, 0.123]	0.235 (0.126) [0.139, 0.214, 0.307]	0.637 (0.126) [0.565, 0.658, 0.732]	0.094 (0.445) [-0.205, 0.101, 0.376]	0.044 (0.194) [-0.061, 0.062, 0.168]
Small firms	0.078 (0.026) [0.059, 0.078, 0.099]	0.254 (0.133) [0.153, 0.235, 0.335]	0.617 (0.133) [0.536, 0.636, 0.718]	0.174 (0.429) [-0.125, 0.177, 0.462]	0.059 (0.186) [-0.049, 0.076, 0.180]
Large firms	0.113 (0.011) [0.105, 0.112, 0.119]	0.187 (0.089) [0.118, 0.181, 0.238]	0.689 (0.089) [0.638, 0.695, 0.757]	0.034 (0.387) [-0.214, 0.016, 0.257]	0.075 (0.170) [-0.025, 0.075, 0.175]
Low-digital industries	0.106 (0.035) [0.078, 0.105, 0.133]	0.225 (0.132) [0.125, 0.200, 0.297]	0.647 (0.132) [0.575, 0.673, 0.747]	0.095 (0.480) [-0.226, 0.088, 0.396]	0.048 (0.207) [-0.061, 0.072, 0.178]
High-digital industries	0.078 (0.020) [0.061, 0.076, 0.094]	0.249 (0.110) [0.169, 0.234, 0.314]	0.624 (0.110) [0.560, 0.640, 0.704]	0.140 (0.383) [-0.114, 0.150, 0.396]	0.044 (0.184) [-0.066, 0.058, 0.167]

Notes: Cells show mean (SD) and [P25, median, P75]. Productivity is centered on the median of the first estimation year in each sample. The materials inversion and output-minus-transitory-shock recovery of Hicks productivity agree to numerical precision after consistent measurement of expenditures, prices, and quantities.

Tables 15 and 16 report IPC distributions. They follow the format of Table 14. The first reports input elasticities and log productivities. The second reports smoothed normalized markups in logs and levels. Standard deviations in these appendix tables describe dispersion across firm-years. They are not bootstrap standard errors.

Table 15: IPC elasticities and normalized productivity distributions

Sample	ε_K	ε_L	ε_M	$\tilde{\omega}_L$	$\tilde{\omega}_N$
Overall	0.039 (0.008) [0.033, 0.039, 0.045]	0.249 (0.134) [0.148, 0.227, 0.325]	0.689 (0.134) [0.613, 0.712, 0.791]	0.122 (0.469) [-0.200, 0.111, 0.425]	0.067 (0.142) [-0.021, 0.057, 0.149]
Small firms	0.034 (0.006) [0.029, 0.034, 0.038]	0.271 (0.142) [0.164, 0.249, 0.355]	0.667 (0.142) [0.582, 0.689, 0.774]	0.157 (0.441) [-0.146, 0.142, 0.435]	0.067 (0.141) [-0.020, 0.056, 0.146]
Large firms	0.042 (0.032) [0.023, 0.045, 0.065]	0.203 (0.097) [0.129, 0.197, 0.259]	0.758 (0.097) [0.702, 0.764, 0.832]	0.016 (0.370) [-0.239, 0.013, 0.256]	0.064 (0.138) [-0.029, 0.051, 0.143]
Low-digital industries	0.037 (0.008) [0.031, 0.037, 0.043]	0.238 (0.140) [0.132, 0.212, 0.315]	0.702 (0.140) [0.625, 0.729, 0.809]	0.138 (0.507) [-0.218, 0.116, 0.476]	0.068 (0.148) [-0.024, 0.055, 0.151]
High-digital industries	0.031 (0.002) [0.030, 0.031, 0.033]	0.269 (0.118) [0.183, 0.252, 0.338]	0.682 (0.118) [0.612, 0.699, 0.768]	0.112 (0.374) [-0.137, 0.125, 0.365]	0.072 (0.128) [-0.008, 0.065, 0.149]

Notes: Cells show mean (SD) and [P25, median, P75] across unweighted firm-year observations; parentheses contain distributional standard deviations. Quartiles use the same median-unbiased convention as Table 14. The overall, small, large, low-digital, and high-digital samples contain 13,291, 9,266, 3,415, 8,278, 4,523 observations, respectively. Estimates use the composite digitalization index and lagged-capital instruments.

Table 16: IPC normalized markup distributions

Sample	$\widehat{m}^\mu$	$\exp(\widehat{m}^\mu)$
Overall	0.067 (0.086) [0.008, 0.046, 0.101]	1.073 (0.104) [1.008, 1.047, 1.107]
Small firms	0.066 (0.085) [0.008, 0.045, 0.100]	1.072 (0.100) [1.008, 1.046, 1.105]
Large firms	0.088 (0.095) [0.027, 0.063, 0.118]	1.097 (0.120) [1.028, 1.065, 1.126]
Low-digital industries	0.068 (0.086) [0.008, 0.047, 0.105]	1.075 (0.101) [1.008, 1.048, 1.111]
High-digital industries	0.081 (0.081) [0.027, 0.061, 0.111]	1.088 (0.095) [1.027, 1.063, 1.118]

Notes: Cells show mean (SD) and [P25, median, P75] across unweighted firm-year observations; parentheses contain distributional standard deviations. Quartiles use the same median-unbiased convention as Table 14. The overall, small, large, low-digital, and high-digital samples contain 13,291, 9,266, 3,415, 8,278, 4,523 observations, respectively. Estimates use the composite digitalization index and lagged-capital instruments. Here $\widehat{m}_{it}^\mu = \ln \mu_{it} - \ln \theta$. Columns report smoothed normalized log markup and its exponential, the markup relative to the common scale θ .

B Robustness Check Using Lagged Materials Demand as the Hicks-Neutral Proxy

The sensitivity specification uses a materials-demand proxy for lagged Hicks-neutral productivity. Its markup-forecast correction is an additional construction. It begins with the labor-productivity block, estimates the markup forecast next, and then estimates Hicks-neutral technology. This order differs from the main IPC sequence. The production technology and markup definition are maintained.

B.1 Materials-demand inversion under IPC

Malikov et al. (2024) extend their method to IPC by using materials demand as a proxy for lagged unobserved Hicks-neutral productivity. We adopt the same proxy approach in this robustness check.

Let $w_{it}^M \equiv \ln W_{it}^M$, $p_{it} \equiv \ln P_{it}$, and $\mathbf{m}_{it} \equiv \ln \mu_{it}$. The materials first-order condition can be written in logs as

$$w_{it}^M - p_{it} = \ln \theta - \mathbf{m}_{it} + \ln \varepsilon_{it}^M + \ln F(K_{it}, L_{it}^*, M_{it}) + \omega_{N,it} - m_{it}, \quad (73)$$

where $\varepsilon_{it}^M = \nu SC_{it}^M$ from the share identities. The constant $\ln \theta$ is retained separately from κ_0 ; it is carried explicitly here to keep the derivation transparent. η_{it} does not enter this condition because it affects measured output and revenue, not the productivity state observed by the firm. From equation (18), $\varepsilon_{M,it}^* = \nu SC_{it}^M$, where $SC_{it}^M = 1 - SC_{it}^L$. Define the scale-free materials-demand index

$$\mathcal{M}_{it}^*\left(\nu, \frac{\nu}{\delta}\right) \equiv (w_{it}^M - p_{it}) - \ln SC_{it}^M + (1 - \nu)m_{it} + \left(\frac{\nu^2}{2\delta}\right)(SC_{it}^L)^2. \quad (74)$$

Substitution of the reduced restricted translog technology in equation (19) into equation (73) gives

$$\mathcal{M}_{it}^*\left(\nu, \frac{\nu}{\delta}\right) = \kappa_0 + \beta_K k_{it} + \frac{1}{2}\beta_{KK} k_{it}^2 + \omega_{N,it} - \mathbf{m}_{it} + \ln \nu + \ln \theta. \quad (75)$$

Thus, under IPC, raw materials demand identifies Hicks-neutral productivity net of the log markup. Treating $m_{i,t-1}$ one-for-one as $\omega_{N,i,t-1}$ would impose an incorrect scale and load persistent markup variation into productivity.

The required markup correction follows from the maintained markup block. Because $R_{it} = \theta\nu\mu_{it}^{-1}e^{-\eta_{it}}$, define the scale-free observed log-markup measure

$$b_{it} \equiv -\ln R_{it} = \mathbf{m}_{it} + \eta_{it} - \ln \nu - \ln \theta. \quad (76)$$

Adding a constant to the dependent variable of equation (61) changes only its intercept. Consequently, b_{it} follows the same transition with intercept $\pi_b = \pi_\mu - (1 - \rho_\mu) \ln(\theta\nu)$ and unchanged slopes (ρ_μ, γ_μ) . Because $\ln \mu_{i,t-1}$ is latent and the observed $b_{i,t-1}$ contains $\eta_{i,t-1}$, it cannot serve as its own instrument. We estimate the observed-signal transition using a constant, $b_{i,t-2}$, and predetermined digitalization as instruments, and then form the one-step forecast

$$\widehat{b}_{i,t-1|t-2} = \widehat{\pi}_b + \widehat{\rho}_\mu(-\ln R_{i,t-2}) + \widehat{\gamma}_\mu D_{i,t-2}. \quad (77)$$

The lagged materials-demand proxy for the location-normalized Hicks-neutral state is

$$\mathcal{H}_{i,t-1}^M\left(\boldsymbol{\Theta}, \frac{\nu}{\delta}\right) \equiv \mathcal{M}_{i,t-1}^*\left(\nu, \frac{\nu}{\delta}\right) - \beta_K k_{i,t-1} - \frac{1}{2}\beta_{KK} k_{i,t-1}^2 + \widehat{b}_{i,t-1|t-2}. \quad (78)$$

Under a correctly specified markup transition, $\mathcal{H}_{i,t-1}^M = \kappa_0 + \omega_{N,i,t-1} + e_{i,t-1}^M$, where $e_{i,t-1}^M$ combines the lagged transitory output shock $\eta_{i,t-1}$ and the negative of the markup-forecast error. The constant κ_0 is absorbed by the dynamic intercept below.

B.2 Alternative sequential nonlinear-GMM block

Substituting equation (78) for the lagged state in equation (46) gives the robustness-check residual

$$\begin{aligned} u_{n,it}^M(\boldsymbol{\Theta}) &= y_{it} - \pi_0 - \beta_K k_{it} - \frac{1}{2}\beta_{KK} k_{it}^2 - \nu m_{it} + \frac{\nu}{2}\left(\widehat{\frac{\nu}{\delta}}\right)(SC_{it}^L)^2 \\ &\quad - \rho_N \mathcal{H}_{i,t-1}^M\left(\boldsymbol{\Theta}, \left(\widehat{\frac{\nu}{\delta}}\right)\right) - \gamma_N D_{i,t-1}, \quad \pi_0 = \pi_N + (1 - \rho_N)\kappa_0. \end{aligned} \quad (79)$$

As in the main IPC specification, the location of the recovered Hicks-neutral composite is fixed by setting its first-estimation-year median to zero. This normalization does not recover the primitive production-function parameters.

The composite disturbance is $\xi_{N,it} + \eta_{it} - \rho_N e_{i,t-1}^M$. Relative to the benchmark dynamic production equation, lagged observed output no longer appears directly. The generated markup-forecast error enters instead. Conditional on the labor-block estimate of ν/δ , estimate $(\pi_0, \beta_K, \beta_{KK}, \nu, \rho_N, \gamma_N)'$ by two-step nonlinear GMM using

$$\mathbb{E}[\mathbf{Z}_{n,it}^M u_{n,it}^M(\boldsymbol{\Theta}_0)] = \mathbf{0}, \quad (80)$$

where the baseline instrument vector is

$$\mathbf{Z}_{n,it}^M = (1, D_{i,t-1}, m_{i,t-1}, k_{it}, k_{it}^2, p_{i,t-1}^M - p_{i,t-1})'. \quad (81)$$

Capital dated t is predetermined by assumption. The lagged variable inputs and relative input price instrument the current static choices and the materials-demand state. These six moments exactly identify the six Hicks-neutral/technology parameters conditional on the labor-block estimate of ν/δ .

The alternative estimation is sequential. We first estimate the labor-productivity block using equations (57)–(59). We then estimate the markup transition and construct the forecast in equation (77). Finally, we estimate the Hicks-neutral block in equation (80) and adjust the markup intercept using the estimated technology scale. Each firm-block bootstrap replication repeats this alternative sequence. The estimation samples are the same as in the main IPC comparison. Output and materials price indices are winsorized at the 2.5th and 97.5th percentiles.

B.3 Identification conditions

The robustness check requires an interior materials choice, $\nu SC_{it}^M > 0$, correct measurement of relative materials prices, and price taking in the materials market. The fitted markup forecast must remove the conditional markup component of materials demand, and its forecast error must be orthogonal to the instruments in equation (81). Lagged digitalization must not shift materials demand directly after conditioning on Hicks-neutral productivity, the markup process, and the included state variables. These are additional assumptions rather than consequences of using a lag. If they fail, γ_N combines technological adoption with an omitted input-demand or markup channel.

C Signal Extraction for Hicks-Neutral Productivity and Markups

This appendix describes Stage 4 of the baseline IPC implementation. It produces the smoothed states used in Table 10, Figure 1, and Appendix A. It does not enter the preceding structural GMM estimates. We first derive the univariate markup restrictions and then the bivariate restrictions for the Hicks-neutral and markup signals. Table 17 reports their empirical checks. The decomposition requires additional second-moment assumptions.

C.1 State and measurement equations

Define

$$\widetilde{RR}_{it} \equiv \ln \hat{\nu} - \ln R_{it}.$$

Ignoring first-step estimation error in $\hat{\nu}$ for exposition, equations (61) and (62) imply the state-space representation

$$\text{state:} \quad \tilde{m}_{it}^\mu = \tilde{\pi}_\mu + \rho_\mu \tilde{m}_{i,t-1}^\mu + \gamma_\mu D_{i,t-1} + \xi_{\mu,it}, \quad (82)$$

$$\text{measurement:} \quad \widetilde{RR}_{it} = \tilde{m}_{it}^\mu + \eta_{it}, \quad (83)$$

where $\tilde{m}_{it}^\mu \equiv \ln \mu_{it} - \ln \theta$ is the log markup normalized relative to $\ln \theta$. Because θ is constant, this normalization changes only the transition intercept: $\tilde{\pi}_\mu = \pi_\mu - (1 - \rho_\mu) \ln \theta$. The persistence parameter, the digitalization effect, and the innovation are unchanged. Hence the Stage 3 composite disturbance is

$$u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_\mu \eta_{i,t-1}. \quad (84)$$

Identification of the two components comes instead from their distinct serial covariance pattern.

C.2 Variance identification

Let

$$\sigma_\xi^2 \equiv \text{Var}(\xi_{\mu,it}), \quad \sigma_\eta^2 \equiv \text{Var}(\eta_{it}),$$

with both variances constant within an estimation sample.

Assumption C.1(Second-moment restrictions). The sequences $\{\xi_{\mu,it}\}$ and $\{\eta_{it}\}$ have mean zero and are serially uncorrelated. They are mutually orthogonal at all leads and lags used below. Their variances are finite and constant within the estimation sample. In addition,

$$0 < |\rho_\mu| < 1.$$

The restriction $\rho_\mu \neq 0$ is needed to identify σ_η^2 from the first-order autocovariance. Values of $|\rho_\mu|$ close to zero imply weak variance separation in finite samples.

Proposition C.1(Variance separation). Under Assumption C.1 and the Stage 3 transition,

$$\text{Var}(u_{\mu,it}) = \sigma_\xi^2 + (1 + \rho_\mu^2) \sigma_\eta^2, \quad (85)$$

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1}) = -\rho_\mu \sigma_\eta^2, \quad (86)$$

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-h}) = 0, \quad h \geq 2. \quad (87)$$

Proof. Using equation (84),

$$u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_\mu \eta_{i,t-1}.$$

The variance result follows because the three terms are pairwise orthogonal under Assumption C.1. For the lag-one covariance, the only nonzero cross-product is generated by $\eta_{i,t-1}$, which enters $u_{\mu,it}$ with effect

$-\rho_\mu$ and $u_{\mu,i,t-1}$ with effect one. This gives

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1}) = -\rho_\mu \sigma_\eta^2.$$

At lags $h \geq 2$, the two composite disturbances contain no common transitory-shock term, and all remaining cross-products are zero by Assumption C.1. This gives equation (87). $\square$

Because $\rho_\mu \neq 0$, equations (85) and (86) identify the two disturbance variances:

$$\sigma_\eta^2 = -\frac{\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1})}{\rho_\mu}, \quad (88)$$

$$\sigma_\xi^2 = \text{Var}(u_{\mu,it}) - (1 + \rho_\mu^2)\sigma_\eta^2. \quad (89)$$

The sample counterparts are

$$\widehat{\sigma}_\eta^2 = -\frac{\widehat{\text{Cov}}(\hat{u}_{\mu,it}, \hat{u}_{\mu,i,t-1})}{\hat{\rho}_\mu}, \quad (90)$$

$$\widehat{\sigma}_\xi^2 = \widehat{\text{Var}}(\hat{u}_{\mu,it}) - (1 + \hat{\rho}_\mu^2)\widehat{\sigma}_\eta^2. \quad (91)$$

Equation (87) provides overidentifying specification checks. In particular, the sample autocovariances of $\hat{u}_{\mu,it}$ at lags two and above should be close to zero. A sizable higher-order autocovariance indicates misspecification of the AR(1) markup process, serial correlation in η_{it} , or both.

The variance estimates must also satisfy

$$\widehat{\sigma}_\eta^2 \geq 0, \quad \widehat{\sigma}_\xi^2 \geq 0.$$

Negative sample estimates indicate that the maintained second-moment restrictions are not supported by that sample. In that case, the signal-extraction step should not be imposed mechanically.

C.3 Kalman filtering and smoothing

Given

$$(\widehat{\pi}_\mu, \hat{\rho}_\mu, \hat{\gamma}_\mu, \widehat{\sigma}_\eta^2, \widehat{\sigma}_\xi^2),$$

equations (82)–(83) form a linear state-space model. The filtering and smoothing are carried out separately for each firm and, in the baseline implementation, separately for each consecutive-year spell. This avoids treating a multi-year gap as a one-period transition. If missing calendar years are retained instead, the transition matrix and accumulated state-innovation variance must be adjusted for the length of the gap.

Let $x_{t|s}$ denote the linear projection of $\tilde{m}_{it}^\mu$ using information through period s , and let $P_{t|s}$ denote its mean-squared prediction error. We use diffuse initialization for the first observation of each consecutive

spell. The standard Kalman recursions shown below apply after this initialization step:

$$x_{t|t-1} = \widehat{\pi}_\mu + \hat{\rho}_\mu x_{t-1|t-1} + \hat{\gamma}_\mu D_{i,t-1}, \quad P_{t|t-1} = \hat{\rho}_\mu^2 P_{t-1|t-1} + \widehat{\sigma}_\xi^2, \quad (92)$$

$$K_t = \frac{P_{t|t-1}}{P_{t|t-1} + \widehat{\sigma}_\eta^2}, \quad x_{t|t} = x_{t|t-1} + K_t (\widetilde{RR}_{it} - x_{t|t-1}), \quad (93)$$

$$P_{t|t} = (1 - K_t) P_{t|t-1}. \quad (94)$$

Because the sample is historical rather than a real-time forecasting exercise, we use the Rauch–Tung–Striebel smoother after the forward pass. For $t = T - 1, \dots, 1$,

$$J_t = \frac{P_{t|t} \hat{\rho}_\mu}{P_{t+1|t}}, \quad x_{t|T} = x_{t|t} + J_t (x_{t+1|T} - x_{t+1|t}), \quad (95)$$

$$P_{t|T} = P_{t|t} + J_t^2 (P_{t+1|T} - P_{t+1|t}). \quad (96)$$

The recursion is initialized at the final filtered values $x_{T|T}$ and $P_{T|T}$.

The smoothed predictor of the transitory output shock is

$$\widehat{\eta}_{it} = \widetilde{RR}_{it} - x_{t|T}. \quad (97)$$

C.4 Recovering Hicks-neutral productivity

The main IPC estimator recovers the composite

$$(\widehat{\omega_{N,it} + \eta_{it}})$$

from equations (52)–(53). The signal-extraction step yields the firm-year predictor

$$\widehat{\omega}_{N,it}^{SE} = (\widehat{\omega_{N,it} + \eta_{it}}) - \widehat{\eta}_{it}. \quad (98)$$

C.5 Extension: joint identification of Hicks-neutral productivity and the markup

The preceding subsections treat the log markup as the only persistent state sharing the transitory output shock η_{it} with an observed signal. The main IPC model in Section 4.4 instead has *two* persistent states, $\omega_{N,it}$ and $\ln \mu_{it}$, sharing the same η_{it} through the two measurement equations (53) and (62). This subsection extends Assumption C.1 and Proposition C.1 to that bivariate case, and is the basis for the identification claim made informally in Section 4.4.

Let $u_{N,it} \equiv H_{it} - \pi_N - \rho_N H_{i,t-1} - \gamma_N D_{i,t-1}$ and $u_{\mu,it} \equiv \widetilde{RR}_{it} - \widetilde{\pi}_\mu - \rho_\mu \widetilde{RR}_{i,t-1} - \gamma_\mu D_{i,t-1}$ denote

the two quasi-differenced residuals implied by equations (68)–(69). Direct substitution gives

$$u_{N,it} = \xi_{N,it} + \eta_{it} - \rho_N \eta_{i,t-1}, \quad u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_\mu \eta_{i,t-1}. \quad (99)$$

The first of these is equation (48).

Assumption C.2 (Bivariate second-moment restrictions). In addition to Assumption C.1 applied separately to each of $\{\xi_{N,it}\}$ and $\{\xi_{\mu,it}\}$, the two innovation sequences may be contemporaneously correlated, with $\sigma_{\xi_N \xi_\mu} \equiv \text{Cov}(\xi_{N,it}, \xi_{\mu,it})$ finite and constant within the estimation sample, but are otherwise mutually orthogonal at all leads and lags. In addition, $\rho_N \neq 0$ and $\rho_\mu \neq 0$.

Proposition C.2 (Bivariate variance and covariance identification). Under Assumption C.2,

$$\text{Var}(u_{N,it}) = \sigma_{\xi_N}^2 + (1 + \rho_N^2) \sigma_\eta^2, \quad (100)$$

$$\text{Var}(u_{\mu,it}) = \sigma_{\xi_\mu}^2 + (1 + \rho_\mu^2) \sigma_\eta^2, \quad (101)$$

$$\text{Cov}(u_{N,it}, u_{N,i,t-1}) = -\rho_N \sigma_\eta^2, \quad (102)$$

$$\text{Cov}(u_{\mu,it}, u_{\mu,i,t-1}) = -\rho_\mu \sigma_\eta^2, \quad (103)$$

$$\text{Cov}(u_{N,it}, u_{\mu,it}) = \sigma_{\xi_N \xi_\mu} + (1 + \rho_N \rho_\mu) \sigma_\eta^2, \quad (104)$$

$$\text{Cov}(u_{N,it}, u_{\mu,i,t-1}) = -\rho_N \sigma_\eta^2, \quad \text{Cov}(u_{\mu,it}, u_{N,i,t-1}) = -\rho_\mu \sigma_\eta^2, \quad (105)$$

$$\text{Cov}(u_{N,it}, u_{\mu,i,t-2}) = 0, \quad \text{Cov}(u_{N,it}, u_{N,i,t-2}) = 0, \quad \text{Cov}(u_{\mu,it}, u_{\mu,i,t-2}) = 0. \quad (106)$$

Proof. Equations (100) and (102) restate Proposition C.1 for the Hicks-neutral signal; the argument is unchanged, so we verify only the new cross-equations. Expand $u_{N,it} = \xi_{N,it} + \eta_{it} - \rho_N \eta_{i,t-1}$ and $u_{\mu,it} = \xi_{\mu,it} + \eta_{it} - \rho_\mu \eta_{i,t-1}$: every cross-product involving two different-period ξ 's, two different-period η 's, or a ξ and an η at any lag vanishes under Assumption C.2. For (104), $u_{N,it}$ and $u_{\mu,it}$ share η_{it} with effect 1 in both, contributing σ_η^2 ; $\xi_{N,it}$ and $\xi_{\mu,it}$ are contemporaneously correlated, contributing $\sigma_{\xi_N \xi_\mu}$; and the lagged terms $-\rho_N \eta_{i,t-1}$, $-\rho_\mu \eta_{i,t-1}$ share $\eta_{i,t-1}$, contributing $\rho_N \rho_\mu \sigma_\eta^2$. Summing gives the stated expression. For (105), the only shared term between $u_{N,it}$ and $u_{\mu,i,t-1}$ (respectively $u_{\mu,it}$ and $u_{N,i,t-1}$) is $\eta_{i,t-1}$, entering with effect $-\rho_N$ (respectively $-\rho_\mu$) in the current residual and effect 1 in the lagged one. Equation (106) holds because, at a two-period separation, no ξ or η term is shared under serial independence. $\square$

Identification. Equations (102) and (103) each identify σ_η^2 directly from a single signal's own lag-one autocovariance, exactly as in the univariate case; equations (105) give two further, distinct routes to the same parameter using the *cross*-covariance between the two signals. This is the sense in which having two signals rather than one strengthens identification: σ_η^2 is overidentified, recoverable four different ways, and the requirement that all four estimates agree is a testable specification check in addition to the three lag-two restrictions in (106). Given any consistent estimate of σ_η^2 , equations (100) and (101) identify $\sigma_{\xi_N}^2$ and $\sigma_{\xi_\mu}^2$, and equation (104) identifies $\sigma_{\xi_N \xi_\mu}$.

The requirement $\rho_N \neq 0$ and $\rho_\mu \neq 0$ in Assumption C.2 is the direct bivariate analogue of $\rho_\mu \neq 0$

in Assumption C.1, applied to each signal separately: if $\rho_N = 0$, equation (102) degenerates to $0 = 0$ and no longer identifies σ_η^2 from the Hicks-neutral signal's own dynamics, though σ_η^2 remains identified from the markup signal's own autocovariance in equation (103) provided $\rho_\mu \neq 0$, and symmetrically if $\rho_\mu = 0$. Equality of ρ_N and ρ_μ is immaterial to identification: all four population moment formulas in equations (102)–(105) continue to identify the common variance when $\rho_N = \rho_\mu$, since each moment equation involves only its own ρ , not the difference between the two. For the bivariate system, at least one nonzero persistence parameter is sufficient to identify σ_η^2 from a lag-one own- or cross-covariance; requiring both to be nonzero is a stronger sufficient condition. Equality of ρ_N and ρ_μ does not create a rank failure.

The own- and cross-covariance estimates and their joint equality checks are reported in Table 17. The overall, small-firm, and low-digital results reject the restrictions. Nonrejection in the remaining samples does not validate them; the negative large-firm Hicks-neutral variance estimate is also incompatible with an admissible variance. These checks do not isolate whether disagreement arises from the common-shock restriction, serial dependence, or misspecified transition equations.

As in the univariate case, these moment conditions identify the second moments the Kalman filter requires; they do not by themselves establish that $\xi_{N,it}$, $\xi_{\mu,it}$, and η_{it} are jointly Gaussian. Absent that additional assumption, the smoothed predictions in equation (70) are the minimum-mean-squared-error *linear* predictors of $\omega_{N,it}$ and $\tilde{m}_{it}^\mu$ given the two signals, used for the reported state-space decomposition; they need not equal the conditional means unless Gaussianity is also assumed.

C.6 Empirical common-shock checks

The displayed variance estimates are

$$\begin{aligned}\hat{\sigma}_{\eta,H}^2 &= -\frac{\widehat{\text{Cov}}(\hat{u}_{N,it}, \hat{u}_{N,i,t-1})}{\hat{\rho}_N}, \\ \hat{\sigma}_{\eta,RR}^2 &= -\frac{\widehat{\text{Cov}}(\hat{u}_{\mu,it}, \hat{u}_{\mu,i,t-1})}{\hat{\rho}_\mu}, \\ \hat{\sigma}_{\eta,\times}^2 &= -\frac{1}{2} \left[\frac{\widehat{\text{Cov}}(\hat{u}_{N,it}, \hat{u}_{\mu,i,t-1})}{\hat{\rho}_N} + \frac{\widehat{\text{Cov}}(\hat{u}_{\mu,it}, \hat{u}_{N,i,t-1})}{\hat{\rho}_\mu} \right].\end{aligned}\tag{107}$$

The cross estimate averages the two directional cross-covariance routes. Their equality with the two own-covariance estimates is a population implication of the maintained restrictions. We test the three displayed estimates jointly using the bootstrap covariance of their differences from the markup estimate. We also report the four-estimate check, keeping the two directional cross estimates separate. The overall estimates are 0.00299 from the Hicks-neutral signal, 0.00475 from the markup signal, and 0.00451 from their cross-covariances. The three-estimate equality test has a reference p -value below 0.001. The restrictions are also rejected for small firms and low-digital industries. The large-firm Hicks-neutral estimate is negative, while its equality test is imprecise. The high-digital test does not reject. These findings qualify the common-shock decomposition and do not establish the Gaussian assumption used to calibrate θ .

Table 17: Common-shock variance estimates and equality checks

Quantity	Overall	Small	Large	Low	High
$\hat{\sigma}_{\eta,H}^2$	0.00299 (0.00065)	0.00440 (0.00075)	-0.00666 (0.10524)	0.00269 (0.00073)	0.00463 (0.00232)
$\hat{\sigma}_{\eta,RR}^2$	0.00475 (0.00046)	0.00585 (0.00059)	0.00256 (0.00040)	0.00439 (0.00048)	0.00604 (0.00075)
$\hat{\sigma}_{\eta,\times}^2$	0.00451 (0.00047)	0.00561 (0.00060)	0.00175 (0.00624)	0.00418 (0.00050)	0.00574 (0.00092)
Three-estimate equality p -value	< 0.001	< 0.001	0.959	0.002	0.737
Four-estimate equality p -value	< 0.001	< 0.001	0.992	0.001	0.894
Joint bootstrap replications	499	499	499	499	499

Notes: H and RR use each signal's own lag-one residual covariance. The cross estimate averages the two directional lag-one cross-covariances after dividing each by its corresponding negative persistence parameter, as in equation (107). Parentheses report firm-block bootstrap standard errors. Equality checks are Wald reference tests using the joint bootstrap covariance of differences from the markup estimate. The three-estimate test has two degrees of freedom. The four-estimate test retains both directional cross estimates separately and has three degrees of freedom. Replications with jointly available variance estimates are used without requiring successful smoothing or markup above one. The markup estimate remains the variance used in the reported smoother. Rejection indicates that the common-shock and serial-orthogonality restrictions are not jointly supported; it does not identify which restriction fails.

C.7 Implementation and interpretation

Several qualifications are important. First, the variance decomposition is credible only when the estimated variance components are nonnegative and the higher-order residual autocovariances are small. Second, variance separation becomes weak when ρ_μ is close to zero. Third, the homoskedastic specification can be relaxed, but then the filtering equations must use the corresponding firm- or time-varying disturbance variances. Fourth, the procedure should be applied to consecutive-year spells unless the state transition is adjusted explicitly for gaps in the panel.

Uncertainty in statistics constructed from the recovered states is assessed by a firm-block bootstrap repeating Hicks-neutral estimation, conditional labor estimation, markup estimation, variance estimation, filtering, and smoothing. The reported decomposition is part of the baseline implementation, but its interpretation remains conditional on the covariance restrictions. Under zero-mean conditional Gaussian η , the maintained variance estimate also supplies $\theta = \exp(\sigma_\eta^2/2)$. Without that shape assumption, second moments alone do not identify this exponential expectation.

D Variable Definitions

- **Investment** (I_{it}): The value of current-period investment in equipment goods, excluding buildings, land, and financial assets. Investment made in period t becomes productive in period $t + 1$.
- **Capital** (K_{it}): Real capital constructed by the perpetual inventory recursion and industry-size initialization described in Section 3, using reported real net capital when available and annual depreciation $d_K = 0.15$.
- **Labor** (L_{it}): Total hours worked by all employees of the firm in year t .

- **Materials (M_{it}):** The real value of intermediate consumption, comprising raw materials, energy, and purchased services. Nominal intermediate consumption expenditures are deflated by a firm-specific intermediate consumption price index constructed from price variations reported in the ESEE.
- **Output (Y_{it}):** The real value of goods and services produced, obtained by deflating nominal production values by a firm-specific output price index constructed using the Paasche index formula.
- **Nominal output and the variable-cost ratio:** $P_{it}^Y Y_{it}$ is nominal goods-and-services production, the denominator used in the model's expenditure shares. The symbol R_{it} denotes total labor and materials expenditure divided by nominal output, not nominal sales.
- **Wage ($W_{L,it}$):** The hourly wage rate, defined as total labor compensation divided by total effective hours worked.
- **Output Price Index (P_{it}^Y):** A firm-specific Paasche price index constructed from price variations of output sales reported in the ESEE, following the methodology of Jaumandreu and Lin (2018).
- **Materials Price Index (P_{it}^M):** A firm-specific Paasche price index constructed from price variations of intermediate consumption reported in the ESEE.
- **Labor share ($s_{L,it}$):** The ratio of total labor compensation to total revenue, $s_{L,it} \equiv W_{L,it} L_{it} / (P_{it} Y_{it})$. It is revenue-denominated and therefore differs from the value-added labor share.
- **Materials share ($s_{M,it}$):** The ratio of total materials expenditure to total revenue, $s_{M,it} \equiv W_{M,it} M_{it} / (P_{it} Y_{it})$.

E Digitalization Index

Table 18: Variables used to construct the digitalization index

Dimension	Variable	Definition
Automation	Use of robots	Whether the company uses robots in the production process (Yes/No)
	Use of CNC machine tools	Whether the company uses computer numerically controlled machine tools in the production process (Yes/No)
	Use of computer-aided design	Whether the company makes use of computer-assisted design (Yes/No)
	Use of flexible systems	Whether the company combines machine tools controlled by computer, robotics, and/or CAD through a mainframe computer (Yes/No)
	Use of LAN	Whether the company uses a local area network in its manufacturing activities (Yes/No)

Dimension	Variable	Definition
Interactions with stakeholders	Ownership of internet domain	Whether the company has its own internet domain (Yes/No)
	Ownership of a webpage	Whether the company's web page is stored on its own servers (Yes/No/NA)
	Sales to companies via internet	Availability of a sales system for other companies through internet (Yes/No/NA)
	Sales to final consumer via internet	Availability of a sales system for final customers through internet (Yes/No/NA)
	Purchases from suppliers online	Whether goods or services are purchased through the internet (Yes/No/NA)
Digital-related human capital	External expenditure on IT training	Value of external expenditure on employee training in computing and information technologies (Euros)
IT technologies	Investment in computer equipment	Share of purchases of information processing equipment in total tangible fixed asset purchases (%)
	Hiring of software programming	Whether the company contracts out software-programming services (Yes/No)

Notes: The global index contains the 13 variables shown. ESEE adoption codes are recoded as 1 for Yes and 0 for No; non-response codes are treated as missing before aggregation. Computer-equipment investment is converted from a percentage of total capital-goods investment into an absolute flow before capitalization. Computer-equipment investment and IT-training expenditure are capitalized with 15% depreciation. Quadrennial indicators are carried forward within firms for at most three years. Section 3.1 describes scaling, normalization, and the treatment of missing components at aggregation.